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Aft Wall Injection Configuration Effects on Cavity Combustion and Flame Stabilization

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in the*



June 2026

Declaration of Authorship

I, Ajani MARTIN, declare that this thesis titled, 'Aft Wall Injection Configuration Effects on Cavity Combustion and Flame Stabilization' and the work presented in it are my own. I confirm that:

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- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
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“We may encounter many defeats but we must not be defeated... In fact, it may be necessary to confront defeat, so you can know who you are, what you can rise from, and how you can still come out of it.”

Maya Angelou

Abstract

Aft Wall Injection Configuration Effects on Cavity Combustion and Flame Stabilization

by Ajani MARTIN

LAFAYETTE COLLEGE

Honors in Mechanical Engineering

Supersonic combustion ramjet, scramjet, technology represents a critical advancement in hypersonic propulsion, overcoming the inherent limitations of conventional air-breathing engines such as turbojets, and ramjets. Both of which require the deceleration of incoming air to subsonic velocities, making them inadequate with the supersonic regime. However, at hypersonic velocities, the residence time of air within the engine is reduced to the order of milliseconds, rendering stable combustion exceptionally difficult. To mitigate this, a cavity flame holder is integrated into the combustor floor to generate a low-velocity recirculating region, providing the necessary flame stabilization and ignition propagation. This thesis aims to examine the performance impacts of varying fuel injection strategies from the aft wall of the cavity into the supersonic airstream. The first configuration utilizes fuel injection from three evenly spaced orifices placed along the bottom of the aft wall. The second configuration utilizes a spanwise slot injection, featuring a continuous slit integrated along the floor. All experimental trials were conducted within the Lafayette College Hypersonic Laboratory shock tube. Qualitative data was acquired using two optical systems: a schlieren imaging setup and a high-speed chemical luminescence camera. Schlieren photography allowed for the visualization of fluid density gradients, allowing for the observation of fuel and air interactions, while the high-speed chemical luminescence camera captured the combustion process for further analysis. These images were subsequently processed in MATLAB to quantify the intensity and spatial distribution of the combustion within the cavity. This quantitative analysis provided the basis for determining the relative performance and efficiency of the two fuel injection configurations. Qualitative analysis showed adequate flame retention and stabilization using the three orifice injection method, at an injection pressure of 80 psi. Spanwise slot injection showed less favorable results due to the over fueling within the cavity. Quantitative results produced through a MATLAB image processing script, showed a cavity burning percentage of approximately 29 % and 42 % for three orifice and slit injection respectively.

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And to all Lafayette students who may be tackling research in Launch Laboratory, know that you know more than you think you do. Progress takes time, and mistakes are an unavoidable part of the process, but both are essential in learning, growing, and ultimately succeeding. Keep pushing forward. You've got this!

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Symbols

A	Area
P_0	Stagnation pressure
T_0	Stagnation temperature
γ	Specific heat ratio
R	Gas constant
M	Mach number
$\dot{m}$	Mass flow rate
ρ	Density
p	Pressure
μ	Viscosity
x	Injection plate length
y	Injection plate width
z	Injection plate height
κ	Permeability
$\bar{d}$	Diameter of solids
ϵ	Volume fraction of pore space
U	Velocity
L	Cavity length
W	Cavity width

*To my friends and family whose support inspires me to strive for
my very best. . .*

Chapter 1

Introduction

1.1 Motivation

Fast and efficient air travel has been a major goal for aerospace since the first flight of the Wright Brothers, who invented the first working airplane. Since the invention of the airplane, engineers have continuously pursued advancements aimed at increasing the maximum operating speeds of aircraft through improvements in aerodynamics, propulsion systems, and structural design. However as it stands there is no consistent way of propelling a plane safely and effectively past supersonic and hypersonic regimes. This is mainly due to the complexities that come with propelling aircraft at these speeds.

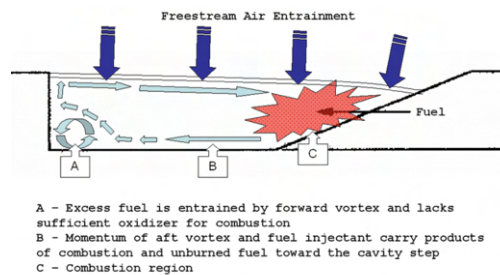


FIGURE 1.1: A drawn representation of the fluid dynamics within a cavity flame holder [1]

Predominantly being how to sustain consistent combustion within the engine to allow for efficient thrust. Modern aircraft propel themselves with turbojet engines, which use rotating components inside of their engines to compress inflowing air before mixing with combustible gas to create propulsion. However, around Mach 3 to Mach 4 the air following into the turbojet engines is moving far too fast for the components of the engine to maintain stable compression. To combat this flaw of the turbojet engine, engineers designed the ramjet engine. This engine uses the overall geometry of the engine to compress the air flowing into the engine by using internal shock waves created by the freestream air [2]. However, this method of experimental engine is only suitable to about Mach 5 to 6, due to the fact that the air needs to be slowed down to subsonic flow before reaching the combustor. To combat this the next stage was creating a scramjet engine, which stands for supersonic combustion ramjet engine. This engine type varies from the ramjet engine because instead of having to slow the incoming air to subsonic speeds, it is able to operate with supersonic air flowing into the combustion chamber. This is due to a recessed area inside of the engine that allows air to shear into the cavity creating recirculation zones that help to hold and slow down the air. This recessed region known as a cavity flame holder will be the main experimental focus of this thesis.

The cavity flame holder has proven to be an effective method for initiating, propagating, and sustaining combustion within scramjet engines. This effectiveness arises from the favorable fluid dynamic environment created by the recessed geometry of the cavity as seen in Figure 1.1. In particular, the cavity increases the local residence time of the flow, promotes the generation and transport of reactive chemical radical elements, and establishes a stable system of recirculation zones within the cavity region. The recirculation zones are crucial because they trap a portion of the high-speed incoming flow, effectively creating a low-velocity region that behaves like a flame stabilization pocket. As

the supersonic freestream air enters the cavity shear layer, it is entrained and partially decelerated, allowing for enhanced mixing between the incoming oxidizer and the fuel injected from the aft wall. The increased residence time ensures that chemical reactions have sufficient time to develop despite the overall high-speed flowing air. Additionally, the cavity promotes radical pooling, chemical molecules accumulate and are stored in the recirculating flow. These radicals are essential in sustaining continuous reactions, thereby lowering ignition delay times and improving flameholding capability under supersonic conditions. The shear layer that forms over the cavity further enhances mixing by continuously feeding fresh reactants into the recirculation zone while also transporting partially reacted gases downstream, reinforcing a self sustaining combustion process. This continuous process can be compared to the effects of a pilot flame, in which the cavity flame holder establishes and maintains the environment necessary to hold materials needed for continuous combustion.

1.2 Previous Work

Previous cavity flame holders investigated the relationship between injection location, ease of and stability of combustion within the cavity flame holder. This included understanding the fluid flow dynamics that are present within the cavity. Predominantly altering the length and depth of the cavity changes the cavity's fluid flow dynamics [3]. The ratio between the length and depth of the cavity dictates whether the cavity operates as an open or closed cavity. The length to depth ratio also dictates the residence time of the cavity and the formation of recirculation zones. The residence time of the cavity is the available time that fuel and air have to mix before they combust. Recirculation zones within the cavity are also vital due to the fact that they store the

necessary materials for combustion to take place. This geometrical difference determines how and where the shear layer from the supersonic freestream air will reattach to the cavity. This study shows that the optimal location for fuel injection is highly dependent on the overall geometry of the cavity.

A previous student, Zev Granowitz conducted a similar experiment for his thesis. Zev conducted research with understanding the effects of floor fueling had on flame stabilization through the use of a cavity flame holder. He conducted three methods of floor injection. Three small injection jets, one singular injection jet, and the last method was using a diffusion plate at the bottom of the cavity. Zev found that the best method of fuel injection was that of the three hole injection. This allowed for adequate shear layer penetration by the fuel, allowing for stable and constant combustion [4]. Harry Chernak, another previous student, designed method of injection, was through a diffusion plate from the floor of the cavity. He found that the biggest problem that he was encountering was managing a way of sealing the fuel in the injection cavity that was located underneath the diffusion plate. He had found that using a gasket was the best method of sealing off the section of the cavity, though he still had minor leakage [5]. Knowledge of both students was used to help the experimental process of this thesis.

1.3 Structure

This thesis presents an overview of the relevant background and theoretical concepts, a detailed explanation of the experimental setup and design methodology, an analysis of the experimental results, and concluding remarks including recommendations for future work. Chapter 2 provides the foundational theory behind scramjet operation, as well as the theoretical framework necessary to support the hypotheses and analyses presented

throughout this thesis. Chapter 3 includes a detailed description of the experimental methods used to support this thesis, including the understanding of testing apparatus and data acquisition tools. Chapter 4 includes a detailed description of the design and manufacturing of the cavity flame holders that were used to answer the main research question. Chapter 5 provides experimental results and analysis of results, using theory explained in Chapter 2. Chapter 6 summarizes the results of testing and provides suggestions for future research. Appendix A provides all relevant drawings to the design process.

Chapter 2

Theory

2.1 Scramjet Engines

Scramjet engines are the next iteration of ramjets and will be essential for the future of super/hypersonic air travel. Scramjet engines were proposed by Maurice Roy in 1946 and verified by Antonio Ferri in 1958. These experimental engines have been praised as the next step needed to achieve supersonic travel as they have built upon the shortcomings of other airbreathing engines.

2.1.1 Turbojet Engines

Turbojet engines are those that are found on commercial and passenger aircrafts. They use rotating mechanical parts within the engine to compress the air flowing into the engine. This fresh air is then mixed with fuel, ignited, and expelled out of the rear of the engine creating propulsion for the aircraft. However, turbojet engines aren't efficient beyond Mach 1 due to the rotating components not being able to keep up with the velocity of the incoming air.

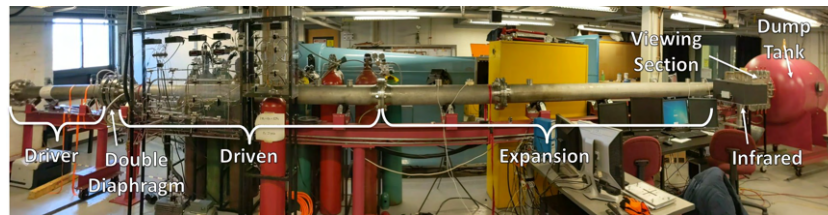


FIGURE 2.1: A labeled image of the Lafayette College expansion tube [6]

2.1.2 Ramjet

Ramjet engines solve the shortcoming of the turbojet engine by eliminating the rotating mechanical components. Ramjet engines use the overall geometry of the engine to compress the flow of air entering the engine. This is done by creating shockwaves within the engine which slows down the air flow to subsonic speeds. Once at these speeds, fuel is then mixed with the fresh air creating propulsion for the engine. The major shortcoming of the Ramjet engine is the fact that the supersonic air flowing into the engine must be slowed down to subsonic speeds. This limits the engine to around Mach 5.

2.1.3 Scramjet

Scramjet engines solve the downfall of the ramjet engine by incorporating a recessed region inside of the engine. This recession is known as the cavity flame holder and will be the focal point of this thesis. It allows the scramjet engine to operate with supersonic airflow throughout the engine unlike the ramjet engine. Like the ramjet engine, the scramjet also uses the geometry of the engine as a means of compressing the air flowing into the engine. Hence giving the scramjet engine the name of supersonic combustion ramjet engine.

2.2 Expansion Tube

All Lafayette College thesis students conducted their research in the Lafayette College Launch Laboratory. This Laboratory in the basement of The Acopian Engineering Center and hosts a 40 foot long Expansion Tube. This experimental apparatus is capable of simulating supersonic speeds, allowing researchers to conduct multiple types of hypersonic research, including supersonic flow imaging and supersonic combustion study.

The expansion tube operates by filling specific sections with highly pressurized gasses which then rush down the length of the tube simulating supersonic flows. The expansion tube is made of 4 main sections, the driver, double diaphragm, driven, and expansion section as seen in Figure 2.1 from the left to the right, which are all sectioned off by aluminum diaphragms. The first section of the tube being the driver holds Nitrogen which is used to accelerate the test gas down the length of tube. The next section of the tube is the double diaphragm section which holds about half of the pressure of the driven section. Upon initiation of the test the gas in the double diaphragm section is quickly vented outside of the tube creating a large pressure difference between it and the driven section, rupturing the aluminum diaphragms. The next section is known as the driven section which holds the test gas. In this thesis the test gas is a mixture of argon and oxygen (20% oxygen and 80% argon), this gas mixture holds oxygen which acts as the oxidizer for combustion. The test gas is propelled down the expansion tube by the driver gas into the test section where the oxidizer should theoretically combust due to its mixing with the fuel as well as the freestream parameters within the tube. The expansion section, which in the case of this thesis was not used, meaning the expansion tube operated in shock tube. This means that the expansion section and the driver section function as one section. The final section of the expansion tube is the viewing

section. This is where the researcher experiment is attached to the expansion tube. This is also where the injection system for the thesis is connected to the main testing apparatus.

The injection system that is held within the test section is connected by a steel braided hose which connects to a bottle of ethylene held on the outside of the test section. Ethylene is used as the combustion gas for this thesis and will not combust when mixed with oxygen and the parameters of the freestream air caused by the reaction of the driver and the driven section.

To begin the experiment the Expansion Tube is vacuumed of all air that resides in it. This process preps the tube for the infilling of necessary gasses that will be used to conduct the experiment. Once the tube has been vacuumed the driver, double diaphragm, and driven section are filled with the to the desired amounts as explained previously. Changing the amount of gas that is within each section allows for the altering of freestream parameters.

Once the driver and driven section have been filled to the desired amount. The test is ready to begin. The injection switch is turned on for a brief second, quickly filling the cavity with fuel. The double diaphragm switch is then turned on which vents the gas with the double diaphragm out off the Expansion Tube. This creates a significant pressure difference between the gas in the driver section and the gas within the double diaphragm, rupturing the diaphragm propelling the driver gas into the test gas into the test section. Where the ethylene will mix with the oxygen starting a chemical reaction igniting the ethylene.

2.3 Flame Holding

Flame holding is the ability of a combustion system to anchor and sustain a flame in a high speed flow, preventing it from being blown downstream by the incoming air. In scramjets, where flow velocities are supersonic and residence times are extremely short, maintaining a stable flame is a notorious challenge. Flame holders, such as the cavity flame holder, are used to create localized regions of low velocity recirculating air [7]. These regions further trap hot combustion products and active chemical radicals, which continuously ignite the incoming fresh fuel and air mixtures. Furthermore, the cavity flame holder can be explained as this pilot flame for the scramjet, holding the necessary elements needed to both propagate and sustain a flame.

The effectiveness of flame holding is tied to flow dynamics, chemical kinetics, and mixture conditions. These parameters are altered by engineers to create the ideal flame holding environment, this includes the geometry of the cavity flame holder, fuel used, and injection type and location. An adequate flame holder must provide sufficient residence time, maintain proper temperature and pressure, and support ideal stoichiometric conditions for combustion [8].

2.4 Fuel Injection Configurations

There are three main locations for injecting fuel into the combustion chamber of the cavity flame holder, pylon, floor, and aft wall, each holding its own set of advantages and disadvantages. Pylon injection injects gas from the front of the cavity allowing for the flow of gas to run parallel and with the flow of freestream air. This method allows for the engine to perform at faster Mach numbers due to flow of gas not getting blown

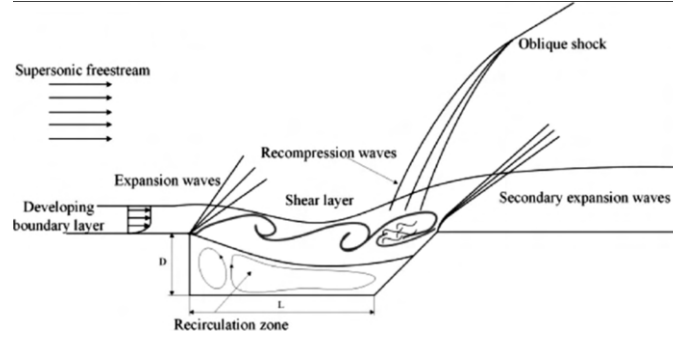


FIGURE 2.2: A labeled diagram of the fluid flow properties of the cavity flame holder [9]

out by the speed and force of the freestream air. However, this method causes major drag restrictions due to the pylons interfering with the flow of the freestream air. Food injection injects the gas from the bottom of the cavity directly into the recirculation zones that form within the cavity. This allows for the added benefits of the gas injecting directly into the area of the cavity where it is needed to reside for stable combustion. However, this method can cause over fueling in the cavity due the main advantage stated previously. Aft wall injection, which will be the main focal point of this thesis, injects the gas from the aft wall of the cavity, parallel and against the freestream flow of air. This method allows for the gas to mix evenly with the air before the air-fuel mixture enters the recirculation zone of the cavity. The main drawback of this method is the possibility of the freestream air blowing out the flame or overpowering the injection pressure of the gas.

2.5 Shear Layer Formation

At the initiation of testing, a supersonic shock wave generated within the expansion tube propagates through the flow field and contributes to raising the local temperature and pressure, acting as a catalyst for the necessary chemical reactions that need to take place within the cavity [10]. As this high speed flow shears over the cavity, this combustion

of shock induced conditions and recirculating flow promotes sustained combustion by stabilizing the flame and maintaining favorable conditions for ongoing reactions.

When supersonic flow passes over the recessed geometry of a cavity flame holder as seen in Figure 2.2, a shear layer forms at the interface between the highspeed external flow and the relatively stagnant fluid within the cavity. This shear layer is generated due to the strong velocity gradient between these two regions and spans the opening of the cavity. As the flow continues downstream, the shear layer becomes unstable and entrains the fluid into the cavity, promoting mixing between the incoming air and any injected fuel. This process is essential in scramjet combustion, as it enhances the interaction between reactants within a very short residence time.

The presence of this shear layer also leads to the formation of recirculation zones within the cavity as seen in Figure 2.2. These zones act as regions of low velocity, trapped flow where hot combustion products and reactive elements can be held, providing a continuous source of ignition for incoming fresh fuel-air mixtures.

2.5.1 Cavity Flame Holder Geometry: Open vs closed cavity

The geometry of the cavity dictates the overall fluid dynamics of the cavity. This means that altering the length and depth of the cavity are two of the most prominent design choices that engineers can make. The ratio between the length and depth of the cavity dictates whether the cavity will function as an open or closed cavity. An open and closed cavity differ primarily in how the shear layer behaves across the cavity and how the internal flow structures are established. In an open cavity, the length to depth ratio is fairly small, 5 to 7, allowing the shear layer to span the length of the cavity without attaching to the downstream wall. This results in a more exposed flow with weaker less

stable recirculation zones, which leads to increased mixing but reduced flame stability. However, a closed cavity geometry allows for more distinct recirculation zones and offers smaller drag penalties. In contrast, a closed cavity has a larger length to depth ratio, 8 to 13 causing the shear layer to reattach within the cavity and form a more confined recirculation zone unlike the open cavity geometry. This in turn creates a much stronger, more stable recirculation zone that is better suited for trapping hot gasses and sustaining combustion, but leads to higher increased drag, making closed cavities generally more effective for flame stabilization in high speed flow environments [11]. This thesis will experiment off of the design of a closed cavity, which will be further explained in Chapter 4

2.6 Stoichiometry Within the Engine

In a scramjet engine the stoichiometry of the chemical reaction between fuel and air is vital due to the inherently small residence times within the engine, which is on the order of milliseconds. Stoichiometry is the quantitative relationship between the amount of reactants used and the formation of products due to a chemical reaction [12]. In terms of thermodynamics and aerodynamics, stoichiometry is most used to find the air to fuel ratio within the engine. This ensures that there is enough oxygen to burn all the fuel that was injected into the air entering the engine.

The stoichiometric conditions within the cavity flame holder are critical for evaluating the efficiency of the various fuel injection methods. Throughout all testing, the oxygen concentration was held constant, while both the injection method and quantity of fuel introduced were varied. This approach enabled the collection of both qualitative and

quantitative data, providing insight into the effectiveness of fuel to oxidizer mixing under the different tested injection conditions.

To provide the necessary conditions for efficient combustion, the stoichiometric balance within the cavity flame holder must be carefully maintained. In this context, stoichiometry refers to achieving the correct ratio of fuel to oxidizer such that sufficient oxygen is present to fully support combustion without either starving the reaction, fuel lean conditions, or overfueling the reaction, fuel rich conditions. Maintaining this balance ensures that the mixture reaches its highest reactivity, where chemical kinetics are most favorable and the energy release is maximized. At or near the stoichiometric ratio of one, the mixture produces the greatest concentration of reactive radicals, which play a critical role in sustaining combustion and stabilizing the flame within the cavity.

This balance also directly influences flame speed and propagation behavior. Flame speeds are maximized at stoichiometric conditions, where the reaction proceeds most rapidly due to optimal radical production and energy availability. Deviations from this ratio introduce reactivity limitations, slowing the flame and potentially leading to instability. For instance, if the flame encounters regions of non ideal mixture, such as pockets of excessive air or fuel, it may propagate into those regions at a reduced rate, diminishing overall combustion efficiency. These effects are often visualized through chemical kinetics plots of reaction rate or flame speed versus stoichiometry, which typically show a peak at the stoichiometric condition.

2.6.1 Mixing of Fuel and Air

Efficient mixing of fuel and fresh air flowing into the cavity is a fundamental requirement for stable and consistent combustion for super and hypersonic airbreathing engines

such as scramjets. This is due to the extremely short residence times associated with supersonic flow, meaning that the fuel and oxidizer must mix rapidly within a limited time and spatial domain. Engineers can increase the cavity's residence time by changing overall geometry of the cavity flame holder, and the location of the injectors which are designed to enhance turbulence within the cavity. Effective mixing helps promote uniform stoichiometric conditions throughout the combustor, which in turn supports consistent ignition and flame stabilization.

The selection of fuel for scramjet operation is also a critical design consideration, as the fuel directly impacts ignition characteristics, combustion efficiency, and overall engine performance. Fuels must fit into a number of categories to be selected as the ideal choice. They must exhibit favorable chemical kinetics, including rapid ignition delay times and high reactivity under supersonic conditions. Additionally, thermal properties such as heat capacity and the ability to undergo endothermic reactions are vital to their performance. Common fuel choices that fit these characteristics are hydrogen and hydrocarbon based fuels. Hydrogen offers fast reaction rates and wide flammability limits, while hydrocarbons provide higher energy density [13].

2.7 Combustion Creation Process

For combustion to occur and be sustained, the cavity flame holder must act as a continuous ignition source. Since the air flowing into the combustion chamber is beyond the speed of sound, a simple ignition spark would not be sufficient, as the engine would need a specific set of fluid dynamics and thermo dynamics conditions to be met to insure that the flame does not get blown out.

The overall purpose of the cavity flame holder is to provide the engine time for the injected fuel and fresh air flowing into the engine ample time for the combustion process to occur. This is done through the formation of recirculation zones within the cavity flame holder. This recirculation zone is where the supersonic air flowing over the open cavity, detaches and creates a subsonic pocket of air within the cavity's recessed area. This is where the necessary components needed for flame propagation and stabilization are entrained. In this case the cavity flame holder can be explained as the pilot flame for the scramjet engine, holding the necessary materials to ensure that a flame is sustained, which is needed for combusting propelling the aircraft forwards.

Radicals are formed through high temperature chemical reactions within the low velocity region of the recirculation zone. Radicals are highly reactive atoms and molecules with unpaired electrons. This makes them highly desperate to bond with anything within reach. Due to this the formation and storing of radicals play a vital role in the sustaining of a flame within a scramjet engine.

Radicals help solve the most difficult part of scramjet propulsion. That being that the air flowing into the engine simply doesn't have enough time to think. Meaning that there simply isn't enough time for it to undergo the initial chemical reactions needed to ignite. However the cavity flame holder solves this through a process called radical seeding. This is done through three steps. The first being the formation of radicals, which are then stored in the hot zones of the recirculation zone. The second step is when the circulating radicals come into contact with the supersonic shear layer flowing over the cavity, where it mixes with fresh air mixing into the engine then with fresh fuel being injected into the engine. The final step is the instant ignition of the fuel air mixture, since the fuel-air mixture has now been saturated, allowing the supersonic air to virtually skip the initial stages of chemical combustion.

2.7.1 Residence Time

Residence time is the amount of time that a fluid particle, containing the fuel air mixture, spends within the cavity before it is swept back into the main supersonic airstream. The entire purpose of the cavity flame holder is to give the air flowing into the engine time to initiate and complete the chemical reactions needed for combustion, which makes controlling the residence time of the cavity vital to the design process. Engineers have the ability to alter the length and the depth of the cavity creating an ideal L/D ratio, which directly dictates the residence time of the cavity. Further allowing for the adjustments of the cavities fluid flow properties.

Chapter 3

Experimental Methods

3.1 Acquisition

The experimental apparatus consists of the, design element as discussed in Chapter 4 Lafayette College Expansion Tube, 5 pressure transducers, and two high speed cameras for Schlieren and CH* chemiluminescence imaging. These various components worked harmoniously to conduct all testing relating to this thesis. Custom parts needed to help support experimental designs were manufactured in the Lafayette College machine workshop.

3.2 Expansion Tube

The Lafayette College Expansion tube is a 40 foot in length 6.3 in inner diameter stainless steel tube that is used to accelerate gasses to super and hypersonic speeds. The tube consists of five main sections [insert an image of the tube with labeled sections]. The first section, from left to right which is the direction of the accelerated gasses, is the driver,



FIGURE 3.1: Image of Schlieren of the cavity flame holder in the Lafayette College Expansion Tube.

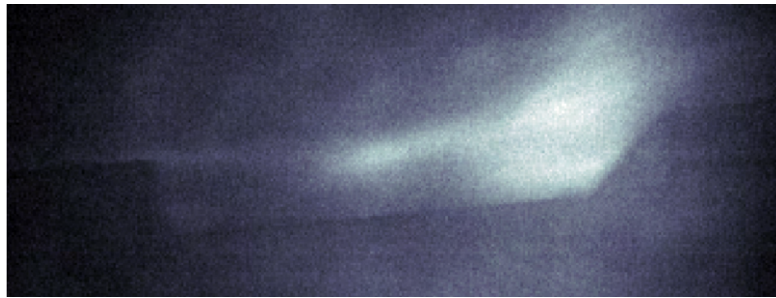


FIGURE 3.2: Image of combustion of the cavity flame holder in the Lafayette College Expansion Tube.

double diaphragm, driven, expansion, and test section. The driver is filled with roughly 750 psi of nitrogen, which is used to drive the test gasses in the driven section of the tube into the test section. The driven section of the tube contains the test gas, a mixture of argon and oxygen. Separating the driver and driven section is the double diaphragm, which consists of two alumine plates that is filled to approximately half of the pressure that is within the driven section. The experiment begins with the rapid depressurization of the space between the two diaphragms, causing the diaphragm to burst. Following the burst of the diaphragms, the gas within the driver section expands into the driven section, creating a shockwave that travels down the tube. The driven section contains the main testing gas that will act as the oxidizer agent needed to initiate combustion within the test section. This gas is forced down the length of the tube by the heavier driven gas during the initiation of the experiment. The expansion and driven section are sectioned off using a plastic diaphragm, allowing for more range within the freestream flow of

gasses through the tube, However, in the case for this thesis the expansion section will not be used, meaning that the expansion section and driven section were not sectioned off. This meant that these two sections functioned as one large driven section. The test section is the location of the cavity flame holder as well as the gas injection system. This is also the location within the tube that combustion will occur and the focal point for the two high speed cameras. The schlierene camera will be used to highlight the differences between the fluid gradients of gasses flowing through the test section [14]. This makes it possible to see various fluid flow properties within the cavity as seen in Figure 3.1. In this image you can see the shock wave passing over the opening of the cavity. The shock wave is the white line that is perpendicular to the floor of the cavity. In this image you are also able to see the shear layers forming from the front edge before the recessed area of the cavity starts. This shear layer is formed due to the different velocities of two parallel flows. These being the hypersonic flow of the air entering the cavity and the slow moving air within the recession of the cavity. Figure 3.1 shows an image taken from the high speed chemical luminescence camera. This image shows the combustion of the fuel and air mixture residing on the aft wall of the cavity.

3.3 Gas Injection System

The combustible gas used for this thesis was ethylene, which holds the similar characteristics to the ideal gasses as described in the previous chapter. This gas is located in a pressurized tank adjacent to the expansion tube. All fitting connecting the steel braided pipes from the gas canister to the test section are Swagelok pipe fittings and all other connections were made with NPT fittings. An Airgas Single Stage Regulator is connected to the bottle of ethylene and controls the amount of pressure that the bottle



FIGURE 3.3: Pressure Transducer location on the Lafayette College Expansion Tube [6]

is releasing into the test section. A 3/8 in. OD, 2 ft. long Swagelok convoluted 316L core stainless steel braided hose connects the regulator to a Swagelok R3A high-pressure proportional relief valve with a maximum pressure of 750 psi. A 3/8 in. 2 ft. long stainless steel pipe connects the relief valve to an ASCO Redhat AC 120V normally closed solenoid valve (8262H023) with a maximum pressure of 275 psi and flow factor, C_v , of 0.35. A second, 3/8 in. OD Swagelok hose connects the valve connects to a steel hose that enters the test section and is bent to connect to a npt fitting that connects the test articles gas injection manifold [4].

3.4 Data Acquisition

The data acquisition system consists of pressure transducers, the two high speed cameras, and supporting devices used to trigger the cameras recording. The pressure transducers along the length of the expansion tube as seen in Figure 3.3 to measure the velocity of the shock wave as well as to trigger the high speed cameras. The five pressure transducers as seen in figure 3.3 are connected to a Kistler Piezotron Power Supply/Coupler Type 5134A Signal Conditioner, whose role is to amplify the signal of the shock wave propagating through the tube using a low pass filter. The signal from the conditioner are then imputed into a National Instruments BNC-2110 BNC adapter that is connected to a NI

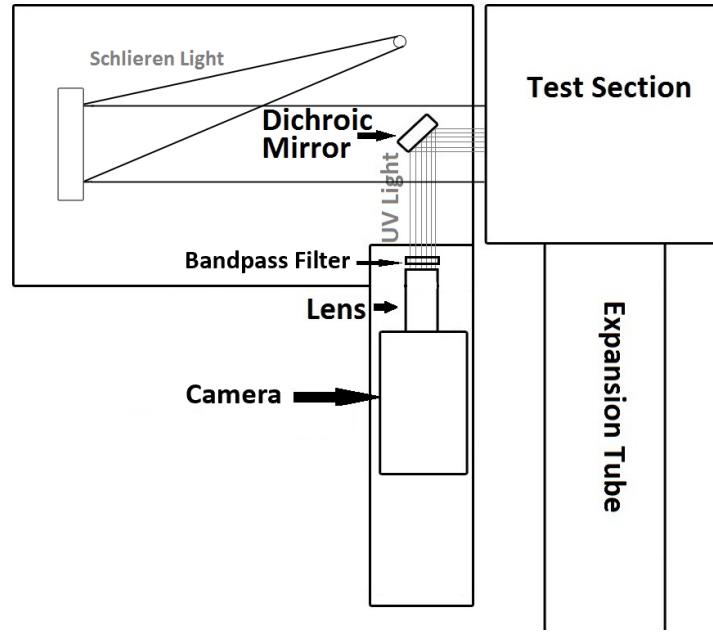


FIGURE 3.4: Simplified Diagram of the CH* Chemiluminescence Camera [4]

PCIe-6346 DAQ. This data is then recorded with a BasicHighSpeedOscilloscope-fixed - PCIe 6346 - trigger LabVIEW code, recording 10 milliseconds at a rate of 0.5 MHz, after being triggered by a rising edge above 0.5V read from the pressure transducers. This signal is also used to trigger the recording of the high speed cameras.

3.4.1 CH* Chemiluminescence Camera

The high-speed camera used for CH* chemiluminescence imaging is a Photron FAST-CAM SA1.1. It is set to record at a resolution of 512 x 256 pixels. The camera setup is as shown in Figure 3.4, with a Chroma T612lpxr dichroic filter (CWL = 612 nm), used to reflect light from the test article to the camera, without affecting the light for Schlieren imaging. The dichroic filter achieves this by transmitting low-frequency light, such as the Schlieren's red LED that operates at 623 nm but reflecting high-frequency light, such as the 430 nm CH* emissions. CH* Chemiluminescence imaging is used to visualize the combustion of ethylene. When ethylene and oxygen react, intermediate

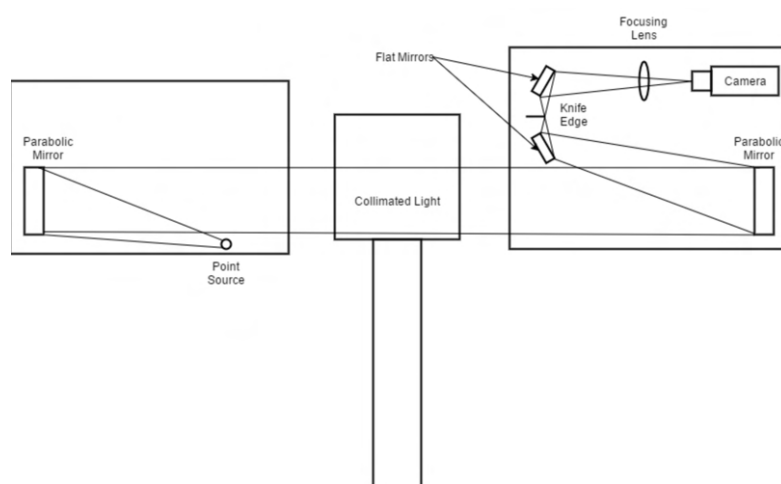


FIGURE 3.5: Simplified Diagram of the Schlieren Camera [4]

reactions cause the formation of CH^* , an excited form of CH . To reach its ground state, CH^* releases a photon that can be visualized at 430 nm. A filter is used that only allows light at 430 nm to pass through, preventing overexposure from other sources of light and ensuring that the light recorded is from the combustion of ethylene.

3.4.2 Schlieren Camera

The high-speed camera used for Schlieren imaging is a Phantom Miro M310, and is set to record at a resolution of 512 x 256 pixels. Schlieren imaging is used to visualize the difference within the density gradients of a flow. To view these gradients, a parabolic mirror is used to create a collimated light source. The light is produced by a Cree 1 Watt Red LED (XRCRED-L1-0000-00M01). This light is passed through the area of interest and the refractive indexes of the gasses within the test section are highlighted due to their differences in density. A series of reflective mirrors as seen in Figure 3.5 direct the light to the knife edge, which adjusts how much light is passed into the lens of the Schlieren camera, adjusting the brightness of the image. Schlieren imaging allows for the visualization of ethylene injected into the cavity, as well as the shear layer impinging on the cavity.

3.4.3 Image Processing

The images from the two high speed cameras were processed in MATLAB. This script was used to crop and rotate the images to get an ideal view of the cavity, allowing for proper analysis of cavity flame engagement. Furthermore a separate MATLAB script was used to test the intensity of the images. This was done by setting a value to each of the pixels, 0 signifying a black pixel and 1 the brightest pixel. This signified where and at what intensity of the cavity is burning. Wheres 0 meant that there was no burning and 1 meaning that there was maximum burning. This process allowed for the qualitative and quantitative analysis of each of the injection methods, showcasing which method produced the most significant and controlled cavity engagement.

3.5 Experimental Procedure

Tests were run using Lafayette College Expansion Tube standard operating procedure. First the Driver and Double Diaphragm sections were unbolted and opened, allowing for the placement of two aluminum diaphragms. The gain for the pressure transcodes were then set and the triggering system was tested to ensure that there were no connectivity issues. The two high speed cameras were calibrated and focused on the test article that was within the test section of the expansion tube. Once all devices were calibrated and set, the tube was vacuumed of all residing air within it, and steel braided lines of any remaining gasses.

Once these intermediate steps were conducted the test was ready to be held. The various sections of the tube were filled with the desired amounts and gasses. Once the tube has been filled with the desired gasses, the injector solenoid valve was activated

approximately 0.5 seconds before initiating a venting of the gas in the double diaphragm section, filling the cavity with fresh fuel. The venting of the Double Diaphragm section causes a large pressure difference rupturing the the diaphragms the seal off the the first three sections of the expansion tube, as explained in [Chapter 2](#)

All sections of the tube were then opened to vent, allowing for the tube to return to atmospheric pressure. The qualitative data from the camera was then collected and analyzed using MATLAB allowing for conclusions to be drawn from results.

Chapter 4

Design Element

The goal of this thesis is to design and manufacture cavity flame holders with different methods of injecting combustible gas into a supersonic airflow through the aft wall of the cavity. The first method is a more conventional method of injection gas into the cavity. This method included 3 small orifices, with an inner diameter of .027 in, located on the aft wall, injecting parallel to the oncoming flow. The second method injection method consisted of a slit spanning, 1.20 in along the middle of the cavity floor floor of the cavity, with a height of 0.003 in. These dimensions for the gas to flow through along with the adjustment of the injection pressure allowed for similar mass flow rates to be matched with the different injection configurations.

The testing of the two methods required the designing of two separate cavity flame holders and gas injection systems, due to the complexities of managing the gas injection with one singular cavity. Each cavity also required the design of its own gas injection manifold

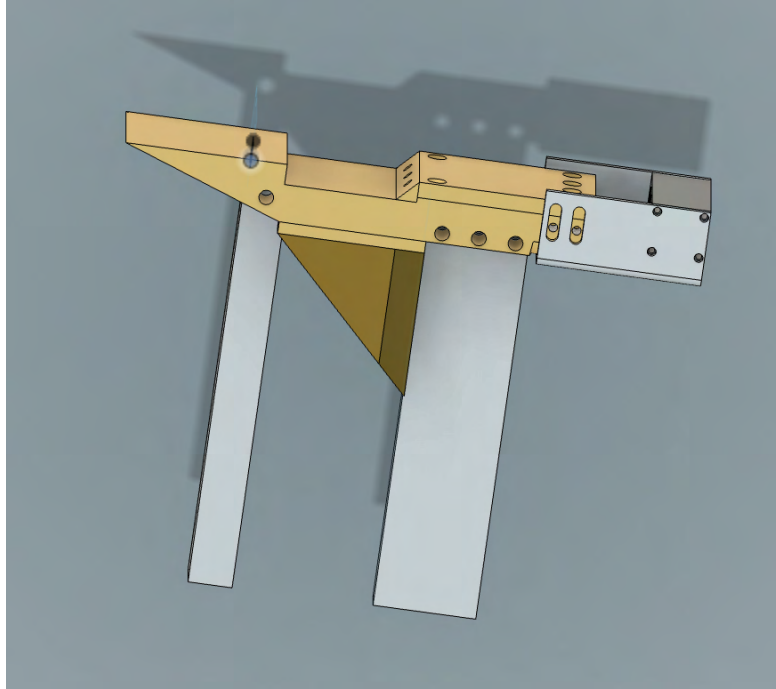


FIGURE 4.1: Three orifice configuration full CAD assembly

4.1 Cavity Design

Both cavities are geometrically the same, the only differences being the gas injection method as gas injection manifold. The constraints of the cavity are fairly minimal as the length, width and height of the entire system is not detrimental to the cavity's function, only being constrained by the size of the test section. However, the geometry of the recessed area of the cavity directly affects the fluid flow dynamics within the cavity as explained in the theory section. Each cavity is 8in in length, 2.5in in width, and 1in in depth, and the aft section of the cavity is angled at a 45 degree angle. The recessed area of the cavity is 1.90in long and 0.35in in depth. This creates a L/D ratio of approximately 5.4, meaning that the cavity operates as a closed cavity. This is vital since closed cavities stimulate stronger recirculation zones which is detrimental to the propagation and stabilization of flames within the cavity.

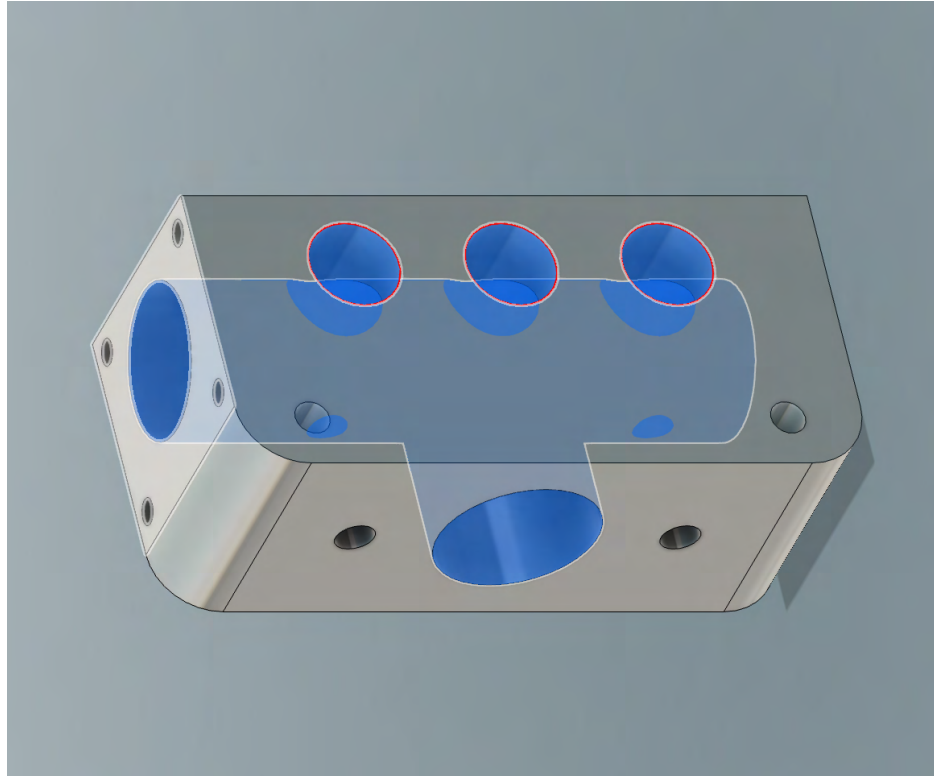


FIGURE 4.2: Three orifice manifold CAD design

To ensure that the cavities were not blown away in the supersonic flow within the expansion tube, two sets of stands were designed to attach the cavity flame holder to the floor of the test section. Figure 4.1 is the initial CAD design of the three orifice injection configuration. This CAD rendering showcases the full cavity assembly, with the gas injection system and support stands connected to the cavity. Figure ?? showcase the model was used for all tests regarding the mentioned configurations and was manufactured in Lafayette colleges machine shop.

4.2 Three Orifice Injection

The three orifice injection was designed with modular intentions in mind to allow for testing of various heights of the injection holes. Through testing it was found that holes locate at the bottom of the aft wall allowed for ideal gas injection. As holes located at

the mid of the aft wall to higher along the aft wall cause the injected gas to inject at a 45 degree angle upwards into the freestream incoming air. This isn't ideal as these methods were injecting gas outside of the cavity rather than in the cavity causing unstable and over combustion. Ethylene ran from steel tubes in the test section to the gas manifold. From the manifold the gas then ran into three small tubes that ran within the rear end of the cavity as seen in Figure 4.2 where it then exited out of the angled aft wall. The flow of gas can be seen by the blue highlights on the previous reference figure.

4.2.1 Three Orifice Injection: Gas Manifold

The gas manifold for the three orifice injection system is a simple manifold design as seen in Figure 4.2. Gas flowing from the test section gathers in the opening of the manifold before it is distributed evenly amongst the three tubes explained above. This method allows for minimal to no gas leakage through the system, confining the combustible gas to the recessed area of the cavity.

4.3 Slit Injection

The slit injection was designed based off of the three hole injection method, but highlighting confining gas from leaking out. The use of O-rings and a special manifold was required since the gas would be flowing along the rear floor of the cavity before entering the cavity, unlike the three hole method where gas was confined to tubes. As seen in Figure 4.3 the blue area highlights where the O-ring runs along the edges of the rear of the cavity before turning 90 degrees downwards along the back of the cavity looping around and running on the opposing side of the cavity. The O-ring is then clamped by a top cover and by the gas manifold to the surfaces of the cavity.

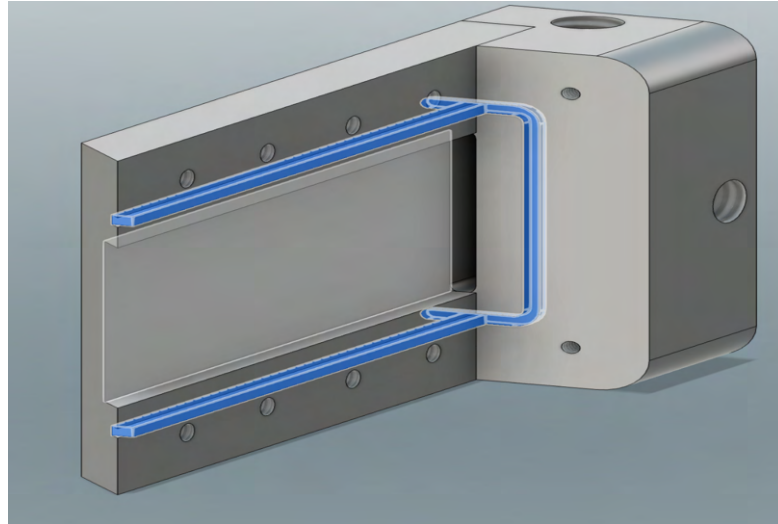


FIGURE 4.3: Design of slit injection manifold O-ring seal

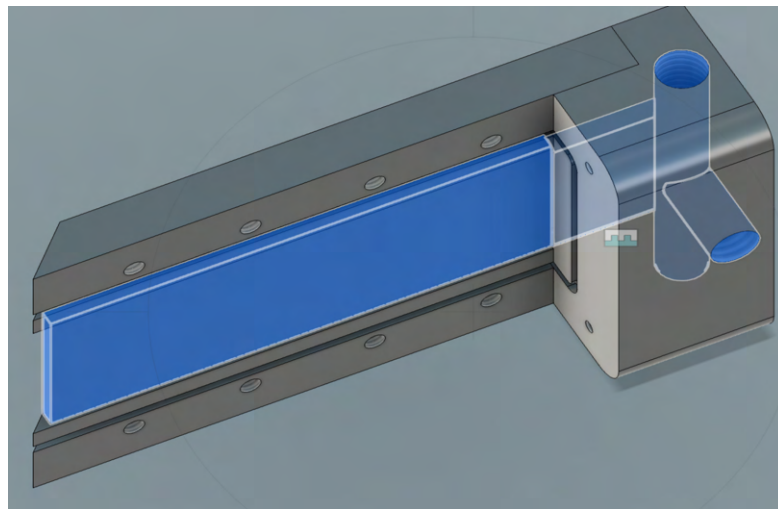


FIGURE 4.4: Design of slit injection manifold

4.3.1 Slit Injection: Gas Manifold

The slit injection gas manifold is slightly more complicated than the previously described manifold due to the concerns stated above, but follows the same concept. Gas will enter the manifold where it is held before being distributed along the rear floor of the cavity, then entering the recessed area of the cavity. The major design difference is the incorporation of O-ring grooves confining the flow of gas within the system before reaching the recessed area. The flow of gas for this configuration is shown in Figure 4.4

4.4 Standards Consulted

The design of wind tunnel models and experimentation in high pressure, high speed wind tunnel systems has several codes and standards that apply. These codes and standards were consulted to ensure that the approach taken in this thesis complied with those methodologies. These include

NASA-STD-8719.28 Wind Tunnel Modeling Systems Criteria - This NASA Technical Standard helps set forth criteria for the design, analysis, quality assurance, and documentation of wind tunnel model systems. It is targeted for particular NASA facilities but can be used to assist in the design of any wind tunnel model.

ASME AED-1 Aerospace and Advanced Engineering Product Definition - This standard was used to ensure the design documentation of the aerodynamic models created for this thesis are consistent with ASME standards.

V V 20 - 2009 (R2021) Standard for Verification and Validation in Computational Fluid Dynamics and Heat Transfer - This standard was used to assist in the determination of experimental error and to compare that with existing CFD simulations of hypersonic cavity combustion.

NFPA 54 / ANSI Z223.1 National Fuel Gas Code, Standard for Installation of Gas Piping and Appliances - This standard was used to ensure that ethylene piping used to bring the fuel in to the test section and cavity assembly was within the safety standards sets out by the NFPA.

NFPA 37 Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines - This standard was used to ensure the combustor hypersonic cavity installation was consistent with NFPA standards.

4.5 Summary of Design Results

Both methods allowed for consistent and reliable injection of combustible gas into the supersonic flow of the freestream air. This allowed for consistent testing for each model producing various qualities of results, dependent on the injection pressure of ethylene into the system.

Chapter 5

Results and Discussion

5.1 Pre-Results Discussion

The goal of this thesis was to design and manufacture two vastly different methods of injecting a combustible gas into a supersonic flow through the use of a cavity flame



FIGURE 5.2: Slit injection test rig

holder. The overall geometries of the cavity flame holder as well as the recessed area of the cavity flame holder were held constant. The cavity parameters are as follows: the cavity is 8in in length, 2.5in in width, and 1in in depth, and the aft section of the cavity is angled at a 45 degree angle. The recessed area of the cavity is 1.90in long and 0.35in in depth. This creates a L/D ratio of approximately 5.4, meaning that the cavity operates as a closed cavity. This L/D ratio is vital as it helps to stimulate stronger recirculation zones which is detrimental to the propagation and stabilization of flames within the cavity. The three orifice cavity is Figure 5.1 and the slit injection cavity is Figure 5.2.

The freestream parameters within the expansion tube as well as the quantity of gas delivered into each section of the tube as explained in Chapter 2 remained constant for the testing of these two models, and the tube was run in shock tube mode. The driver was filled with 700 psi of nitrogen, the double diaphragm was filled with 350 psi of nitrogen, and the driven section was filled with 0.5 psi of a mixture of argon and oxygen (80 % argon and 20 % oxygen). The nitrogen is used solely to accelerate the argon-oxygen mixture to the desired freestream speed needed to conduct the test. These parameters allowed for a freestream speed of around Mach 1 to 1.5 and a temperature high enough for the auto ignition of the gas-air mixture.

5.1.1 Forms of Analysis

This thesis required two forms of data collection and data analysis. The first and most prominent method of analysis was done through the use of qualitative data. As stated before in Chapter 3 and Chapter 2, the two high speed cameras were used to collect the changes in fluid gradient and whether there was combustion within the cavity. Before further analysis was done, each experiment was checked to see if there was adequate

combustion found within the cavity. Tests where there was no combustion were scrapped and noted as to not repeat the parameters used to run that initial test. Once combustion was found the images shown by the high speed chemical luminescence camera were used to determine if combustion was found within the cavity itself, or if combustion occurred outside of the cavity.

After qualitative analysis determined the ideal and preferred testing conditions for a configuration, quantitative analysis was then continued for that specific test. This was done through a MATLAB script, where the recessed area of the cavity was selected from the desired image. The script then gave a number to each of the pixels within the image ranging from 0 to 1. 0 meant that the selected pixel was pure black, showing that there was no source of burning at that location in the image. 1 meant that the selected pixel was pure white, showing that there was maximum burning at that location in the image. This method allowed for the analysis of determining the intensity of burning within the selected image of the cavity.

A second MATLAB script was used to locate the top 5 % of burning pixels gained from the script stated above. These pixels are regarded as where combustion within the cavity is located. They were assigned red dots then highlighted on top of the original image. This allowed for analysis and location of where combustion occurred within the cavity for each configuration tested. Furthermore, a third MATLAB script was used to find the location of all burning pixels, pixels with an intensity of greater than 0, and used this data to create a heatmap showcasing the intensity of burning. Ideally the location of the most intense pixels should be found on the aft ramp of the cavity, signifying that that combustion has occurred within the cavity and stabilized on the aft wall before exiting the system.

These three MATLAB scripts allowed for the proper analysis and sourcing of all qualitative images. This further allowed for a conclusion to be drawn regarding which aft wall configuration is the most ideal for cavity combustion and flame stabilization.

5.2 Three Orifice Injection Method Analysis

Preliminary testing of the three hole configuration determined that locating the holes a $\frac{1}{4}$ inch above the cavity floor produced the most ideal results. Previous testing was done with a hole location a $\frac{1}{2}$ above the floor of the cavity. However, this resulted in the ethylene injecting at a 45 degree angle upwards, above the cavity recession. This led to the fueling of the entire test section rather than purely the cavity, resulting in over combustion occurring.

Testing with the expansion tube parameters explained above, injection pressure of ethylene ranging from 60 psi to 80 psi, and with an aft wall injection height of a $\frac{1}{4}$ inch, resulted in consistent combustion and was used for the analysis explained in the next section. The orifice of a single injection tube had a diameter of 0.027 inches, resulting in an area of 2.29×10^{-3} inches squared. This means that the total area for the injected ethylene to flow through all three tubes is 6.87×10^{-3} inches squared. Simplifying assumptions such as there being no pressure drops from the canister of ethylene, through the steel braided hose and into the test section, and internal friction of the gas injection system were made to keep analytical results as consistent as possible. With these assumptions made the mass flow of ethylene through all three orifices was found to be 1.3596 g s^{-1} .

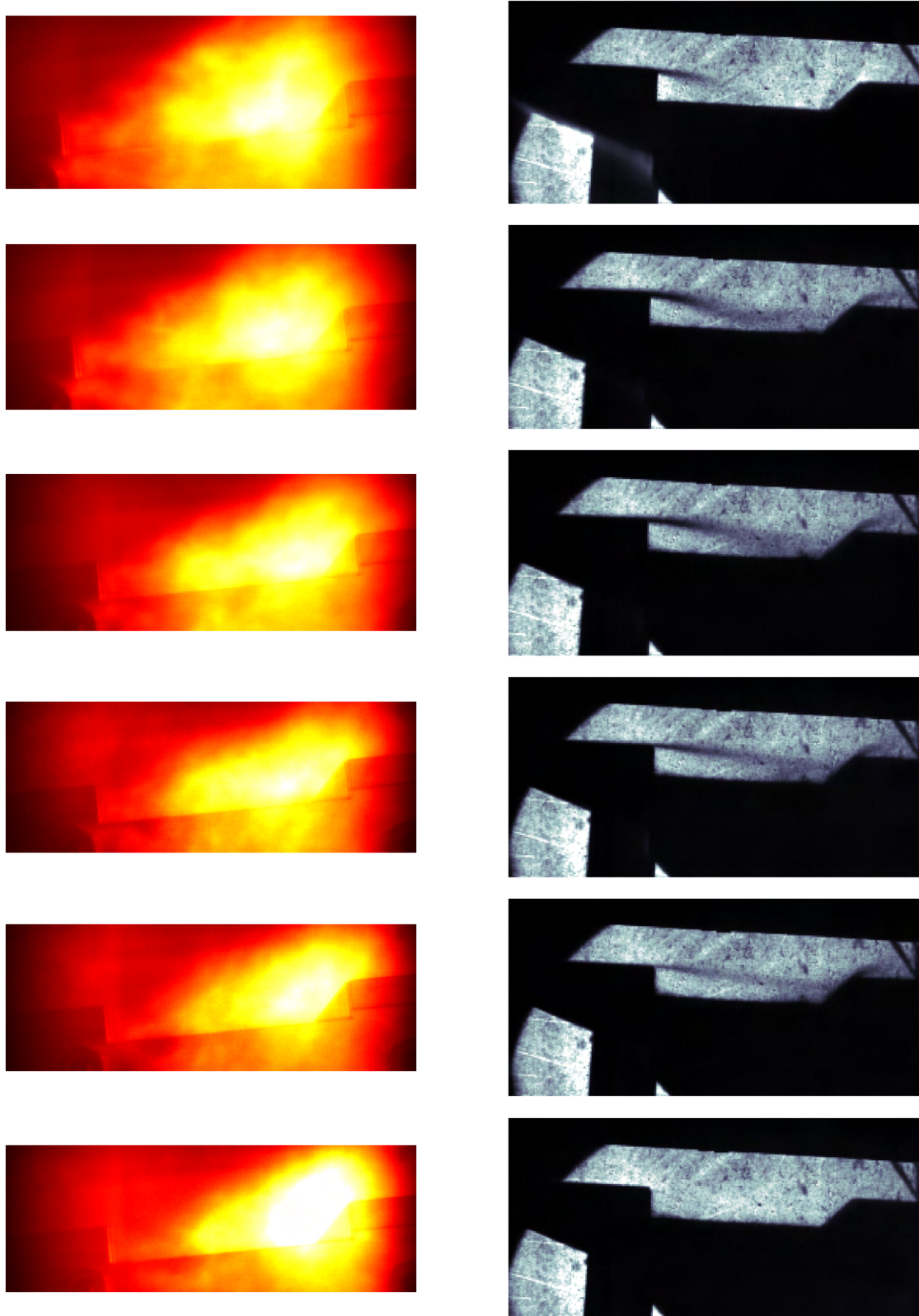


FIGURE 5.3: Combustion (left) and schlieren (right) images of the three orifice injection method [0 microseconds to 250 microseconds]

5.2.1 Qualitative Analysis

The second column of images in Figure 5.3 shows the formation of the fluid shear layer after the initiation of the experiment taken from the schlieren camera. The first image shows the shock wave passing over the rear of the cavity, the shock wave is the bright white line perpendicular to the top of the cavity. This shockwave is a high pressure supersonic wave that dramatically compresses and heats the air- fuel mixture. This leads to a near instantaneous auto ignition of the fuel-air mixture within the cavity. The following images show the formation of the shear layer that dips into the cavity. This shear layer helps separate the supersonic flow of air flowing over the top of the cavity and the slow moving air inside of the cavity. This helps create recirculation zones which are vital to the formation of the necessary element needed for stable combustion as explained in Chapter 2.

The first column of images in Figure 5.3 shows the images preceding the images previously described. These images were taken from the chemical luminescence camera and show the propagation of combustion to the stabilizing of combustion on the aft ramp of the cavity. As seen in the images combustion begins on the floor of the cavity, where recirculation zones are the strongest before traveling to the aft wall where it then stabilizes. Qualitatively, the last image in row 1 of Figure 5.3 is the ideal outcome of testing. This shows that using the three orifice, at a $\frac{1}{4}$ inch height above the cavity floor, with an injection pressure of 80 psi resulted in adequate combustion and combustion stabilization. These parameters were chosen for the quantitative portion of analysis for the method of injection.

5.2.2 Quantitative Analysis

The final image of the left column of Figure 5.3 was the image used for quantitative analysis as explained above. This image was chosen due to combustion seeming to stabilize on the aft wall of the cavity.

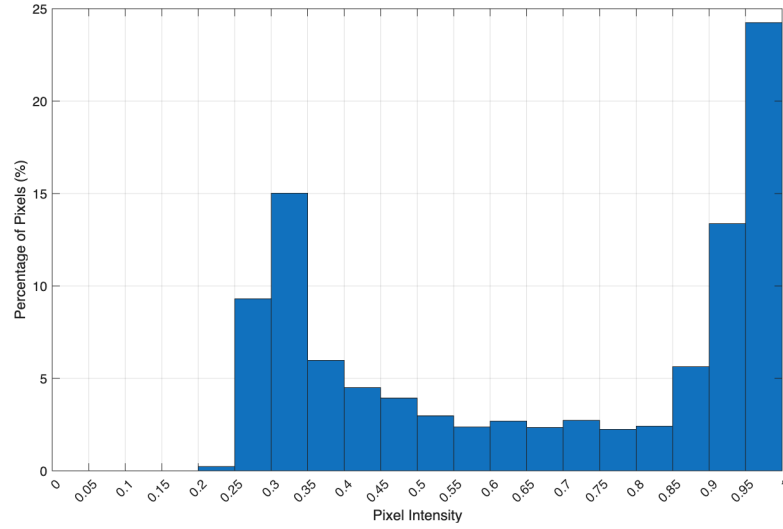


FIGURE 5.4: Percentage of burning pixels vs percentage of pixels with the cavity for the three orifice injection method

Figure 5.4 shows the relationship between the percentage of pixels with the cavity to the burning intensity of those pixels. As explained above the intensity rating ranges from 0 to 1. The Figure shows that there are 0 pixels that are true black. This means that every possible location in the cavity is burning. The Figure shows that majority of the cavity is burning with a pixel intensity of 0.3. This means that almost all of the cavity has experienced some level of combustion. This is due to the recirculation zones within the cavity holding the necessary elements as explained in Chapter 2 needed for the propagation of combustion. There is approximately 24 % of the cavity that is experiencing a pixel intensity of 1. This means that the location of those pixels is where the stabilization of combustion should reside. While Figure 5.4 shows favorable results, it doesn't tell where the location of pixels are within the image. Technically speaking,

the pixels of an intensity of one may be at the front edge of the cavity which would not be ideal and would lead this configuration results to lead to a null conclusion.

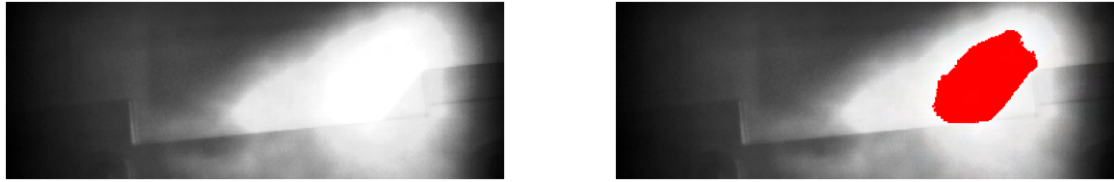


FIGURE 5.5: Original image of cavity combustion (left) localized pixels of burning (right)

Figure 5.5 shows the original cavity image that was used for quantitative analysis in MATLAB. In MATLAB the original image was processed and pixels that experienced an intensity of 0.95 to 1 were marked with a red dot. These pixels were then highlighted and added to the original image. This process allowed for the conclusion to be drawn on whether combustion stabilized within the cavity. The Figure shows that combustion did indeed occur within the cavity and most noticeably, it stabilized on the aft wall of the cavity. At this location the fuel from the injection jets are directly fueling the previous combustion while fresh air is directly applied for further combustion. The result gained from this graph line up with the results shown on Figure 5.4 as that both show that a majority of pixels within the cavity are burning. However Figure 5.5 solidifies that the location of burning is at an ideal location.

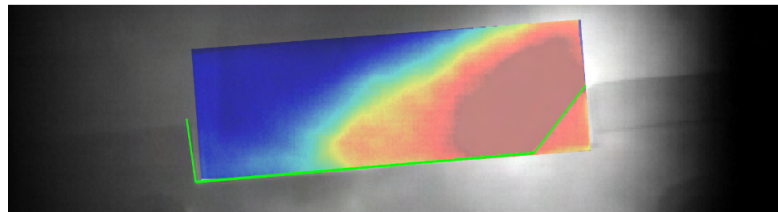


FIGURE 5.6: Heatmap of burning intensity for the three orifice injection method

Figure 5.6 further reinforces the claims stated above by showing an overlaid heat map of the processed area on top of the original cavity image referenced above. The highlighted

green lines show the outline of the recessed area of the cavity. The dark red represents the most intense pixels within the cavity and the blue represents pixels with an intensity of 0.25 and less. This further supports the claims made above by showing that the area of mac burning aligns with the red dotted pixels on Figure 5.5.

5.3 Slit Injection Method Analysis

The cross sectional area of the slit in which ethylene would enter the cavity was 3.6×10^{-3} inches squared, approximately 1.57 times greater than the cross section area of the three orifice injection method. Due to the greater cross sectional area of this method, the pressure out of the ethylene canister had to be decreased as higher pressures resulted in over combustion within the cavity. It was found that 10 psi of injection pressure offered the best results for slit injection. This means that this method is capable of injecting more fuel directly into the cavity and at a more advantageous position within the cavity, being directly on the floor of the cavity. The mass flow of ethylene injected with this method was found to be 14.0229 g s^{-1}

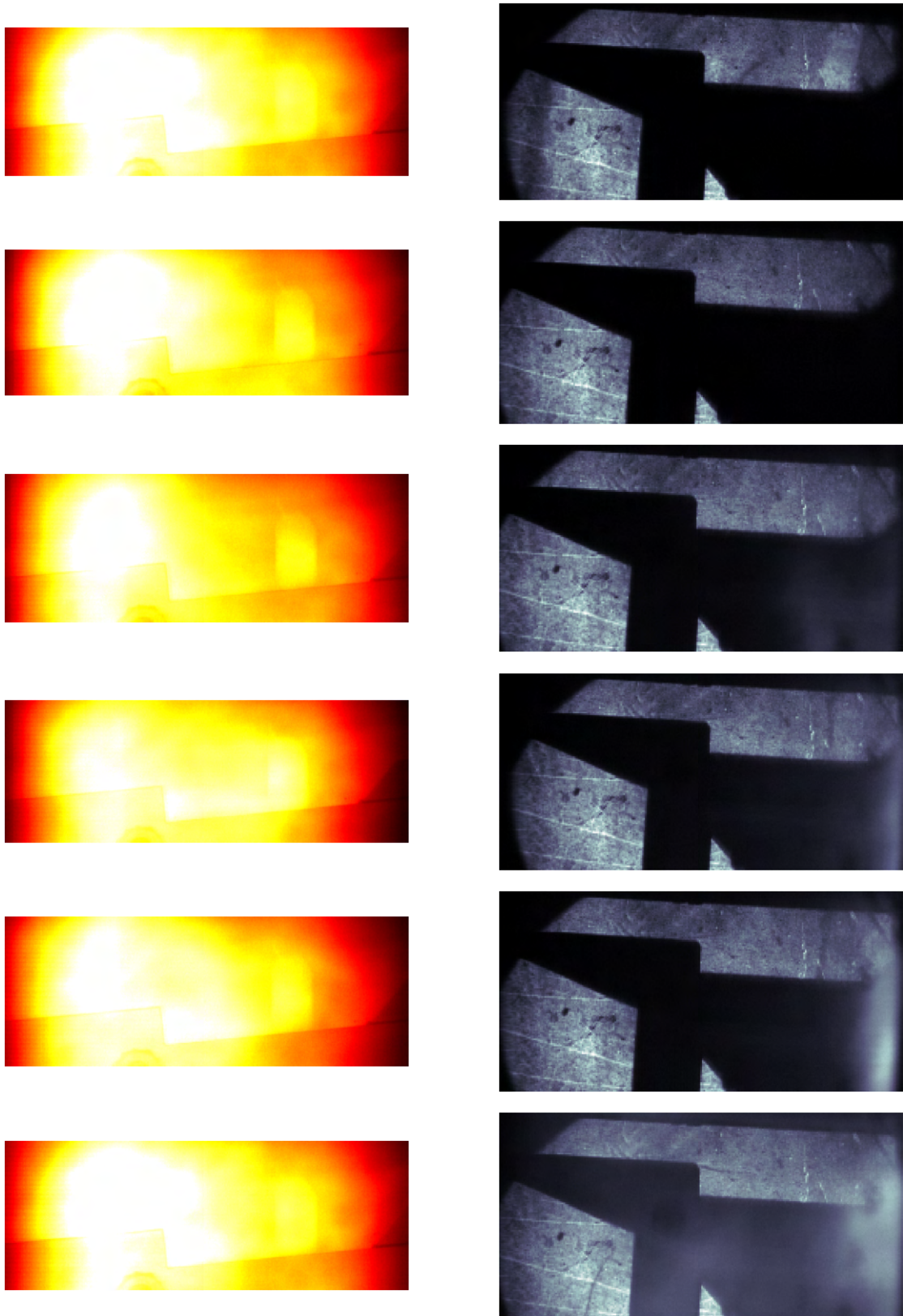


FIGURE 5.7: Combustion (left) and schlieren (right) images of the slit injection method
[0 microseconds to 250 microseconds]

5.3.1 Qualitative Analysis

The second column of Figure 5.7 shows the progression of the shock wave after it passes over the recessed area of the cavity. As the images continue there is a noticeable bright light that starts to creep across the images from the left to the right. This is possible combustion happening outside of the cavity before propagating into the cavity itself. This could be caused by many factors such as the over-fuelling of the cavity. However, in this case it is believed to be due to the ethylene benign blown back by the force of the freestream air. Unlike the three orifice model, the injection pressure at the the opening of the slit was not strong enough to penetrate through the flow of the freestream air.

Upon analysis of the first column of Figure 5.7, combustion propagated on the front of the cavity. This is likely due to the ethylene not having enough inertia to push through the flow of air but inside pooling up on the front of the cavity before igniting due to the free stream parameters. While this is not favorable, this configuration of the slit injection method and 10 psi of injection pressure produced the best qualitative data for further analysis. Instead of the entire cavity combusting only a select region combusted, showing stabilization, even if it wasn't at the ideal location.

5.3.2 Quantitative Analysis

The final image of the left column of Figure 5.7 was the image used for quantitative. This image was chosen due to combustion seeming to stabilize. However, in this case combustion stabilized at the front of the cavity rather than the aft ramp.

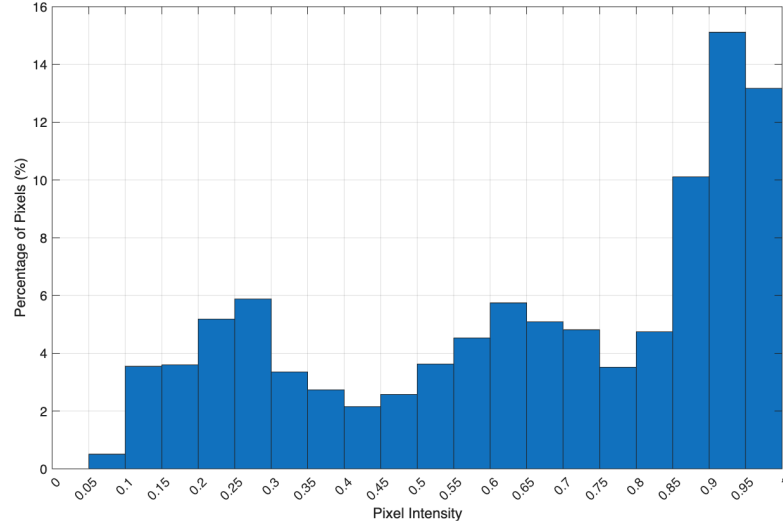


FIGURE 5.8: Percentage of burning pixels vs percentage of pixels with the cavity for the slit injection method

Figure 5.8 shows the percentage of the cavity as well as the top of the front of the cavity that was burning. The figure shows that approximately 13 % of the cavity was burning with a value of 1. This is significantly lower than that of the three orifice injection methods. Furthermore, the increase in pixel intensity throughout the plot signifies that the slit injection model is potentially experiencing over combustion of fuel, which could be due to the higher mass flow of the injection geometry. Furthermore, the decrease in pixels with an intensity rating of 0.5 leads to the conclusion that combustion may have not occurred within the cavity. Instead combustion may have propagated outside of the cavity before igniting fuel that had pooled up at the front edge of the cavity.

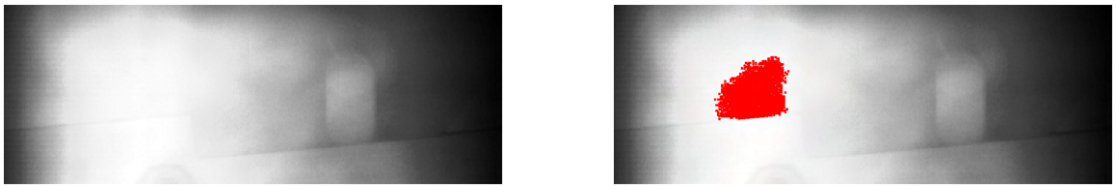


FIGURE 5.9: Original image of cavity combustion (left) localized pixels of burning (right)

Figure 5.9 shows the original image that was used to analyze through the MATLAB

script explained above. The image on the right highlights the pixels with an intensity rating of 0.95 to 1 with red dots. This Figure further helps explain the results gained from the previous figure mentioned. It shows that combustion stabilized on the front wall of the cavity and pooled over to the top edge of the cavity. Upon further analysis of schlieren imaging depicted in Figure 5.7, ethylene had been pushed to the leading edge of the cavity and flowed over the top of the leading edge prior to the shockwave reaching the cavity. This meant that ethylene had over fueled the cavity prior to combusting. This is likely due to the flow of ethylene out of the opening of the slit geometry. Rather than the ethylene injecting as three defined streams of pressurized gas, it is injected as a wave of gas into the cavity pooling towards its leading edge.

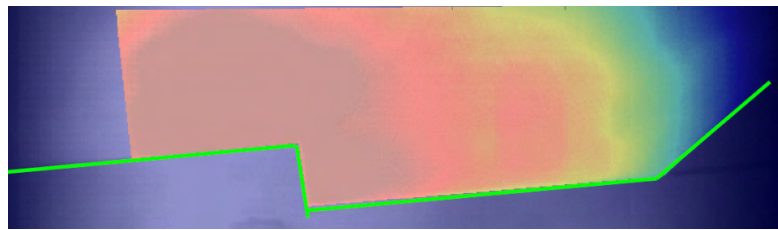


FIGURE 5.10: Heatmap of burning intensity for the slit injection method

Figure 5.10 shows an overlaid heat map of the pixel of different intensities laid over the original image that was used for processing. This figure shows that virtually the entire front $\frac{2}{3}$ was burning with an intensity of 0.8 to 1, while the aft of the cavity experienced little to no burning. This result is the complete opposite of that found with the three orifice injection method. While combustion did stabilize within the cavity, it did not stabilize on the aft wall, leading to the conclusion that slit injection is inferior to that of the three orifice injection method.

5.4 Flame and Shockwave Propagation

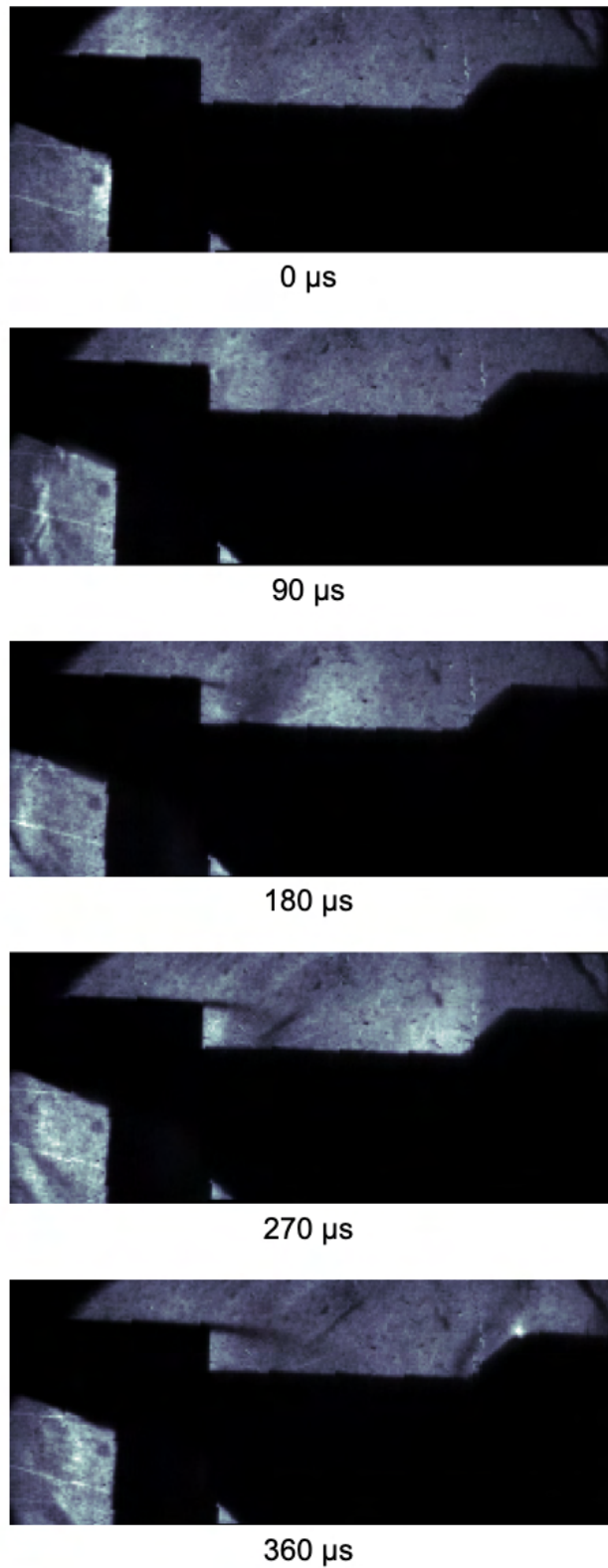


FIGURE 5.11: Shockwave propagation over the cavity for the three orifice injection method

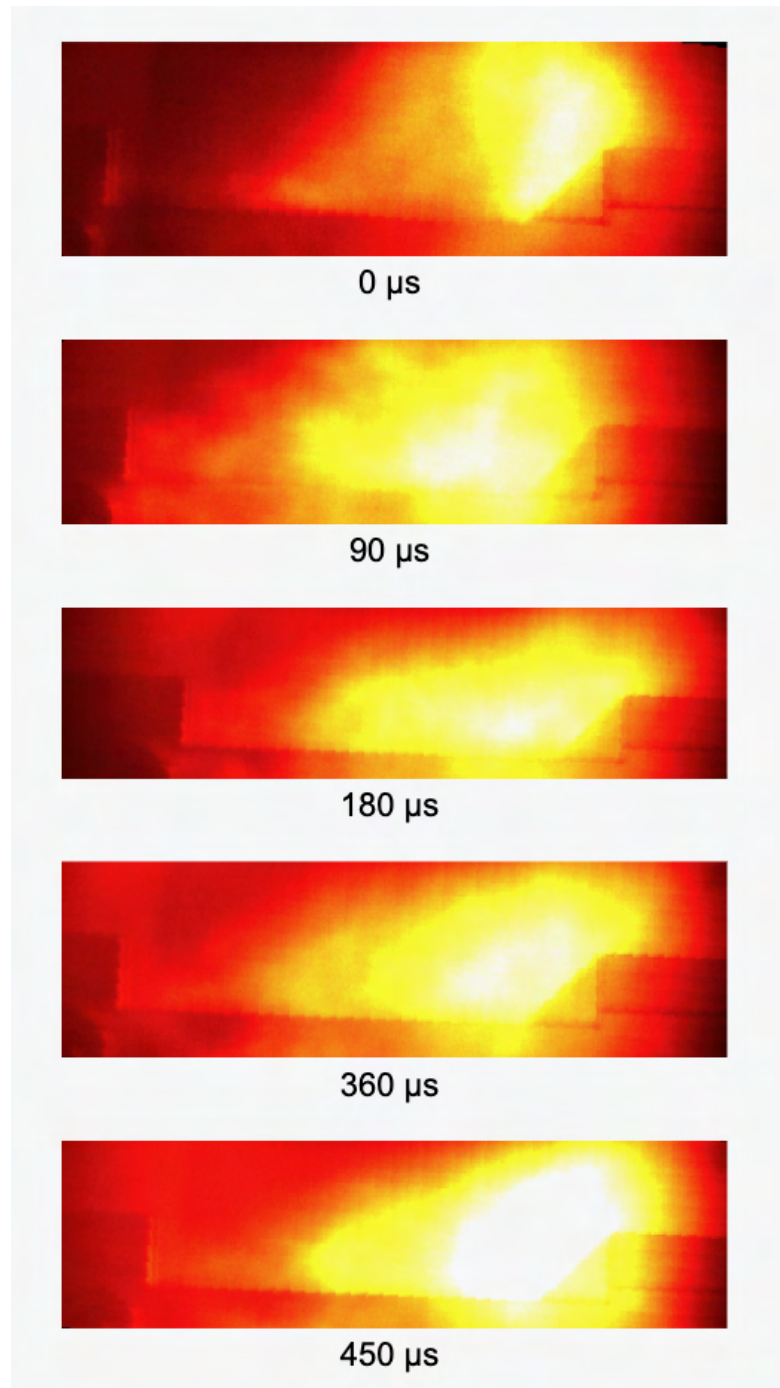


FIGURE 5.12: Combustion propagation over the cavity for the three orifice injection method

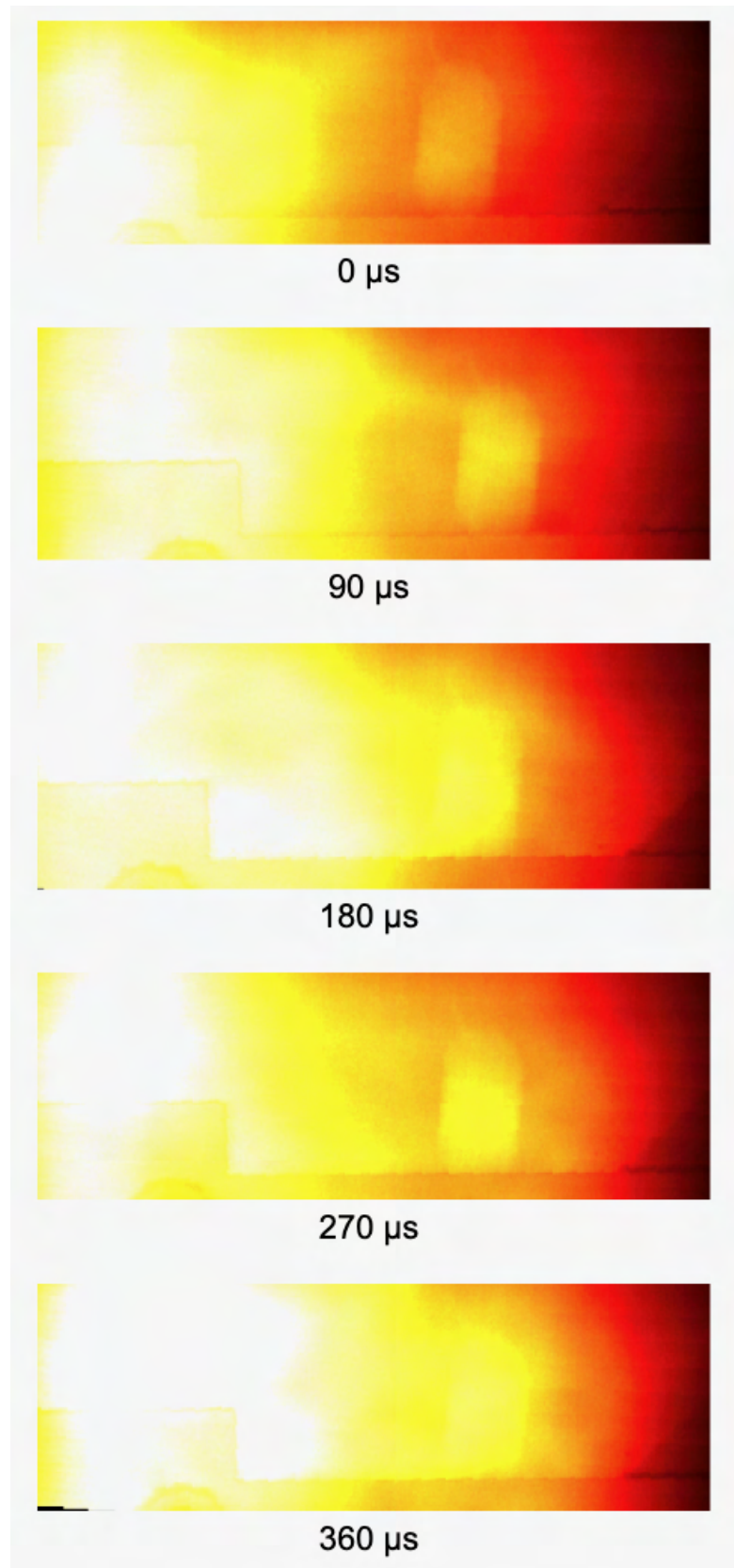


FIGURE 5.13: Combustion propagation over the cavity for the slit injection method

Figures in section 5.4 show the propagation of the shockwave and combustion over the cavity for both methods of injection. The shockwave propagation for the three orifice injection method shows that the shockwave passes over the cavity before combustion occurs. This is ideal as it allows for the shockwave to compress and heat the fuel-air mixture within the cavity before auto ignition occurs. The combustion propagation for the three orifice injection method shows that combustion begins on the floor of the cavity before propagating to the aft wall where it stabilizes. This is ideal as it shows that combustion is being fueled by the fuel from the injection jets while fresh air is being applied to the previous combustion, allowing for stable combustion to occur. The combustion propagation for the slit injection method shows that combustion begins at the front edge of the cavity before pooling up and over to the top edge of the cavity. This is not ideal as it shows that fuel was not able to penetrate through the flow of freestream air and instead pooled up at the front edge of the cavity before igniting.

Chapter 6

Conclusion and Future Work

This thesis sought to find what were the effects of aft wall injection configurations on the ability of a cavity flame holder to propagate and stabilize combustion. This was done through the experimentation of the two configurations. The first configuration injected ethylene through three small orifices located at the bottom of the aft ramp. The second configuration injected ethylene from a slit spanning across the floor of the cavity. All parameters relating to the flow of air in the tube were held constant. This allowed for the formation of a shock wave that would initiate the chemical reaction within each of the cavities and a temperature hot enough to initiate auto ignition of the fuel.

Consistent results for the three orifice injection methods were obtained with an injection pressure of 80 psi which gave a mass flow rate of ethylene of 1.3596 g s^{-1} , through a combined cross sectional area of 2.29×10^{-3} inches squared. These parameters allowed for consistent combustion of the fuel-air mixture and stabilization of the flame on the aft wall of the cavity. Testing of the slit injection method found less favorable results. The best data was obtained with an injection pressure of 10 psi which gave a mass flow rate of ethylene of 14.0229 g s^{-1} , through a cross sectional area of 3.6×10^{-3} inches

squared. This resulted in a 1.57x increase in area and a 10.31x increase in mass flow and from the three orifice to slit injection method.

Quantitative analysis showed that the slit injection method saw a 44.8 % increase in the number of pixels that experienced an intensity rating of 1. This leads to the conclusion that the cavity had become oversaturated with ethylene, destabilizing the stoichiometry of the cavity. This is further supported by the propagation of combustion shown with the chemical luminescence camera of the slit injection method. It can be seen that combustion occurred behind the rear of the cavity before igniting the gass that was pooling on and over the front wall of the cavity. This is also likely due to the oversaturation of ethylene caused by the higher mass flow of ethylene with this method. On the other hand, the three orifice injection method showed favorable results. When analyzing the chemical luminescence camera, it can be seen that combustion occurred within the recirculation zones within the cavity before stabilizing on the aft wall of the cavity. Once it stabilized on the af wall it was feed with fresh fuel directly from the injection location and fresh air flowing from the shear layer that formed due tp the shock wave passing over the cavity.

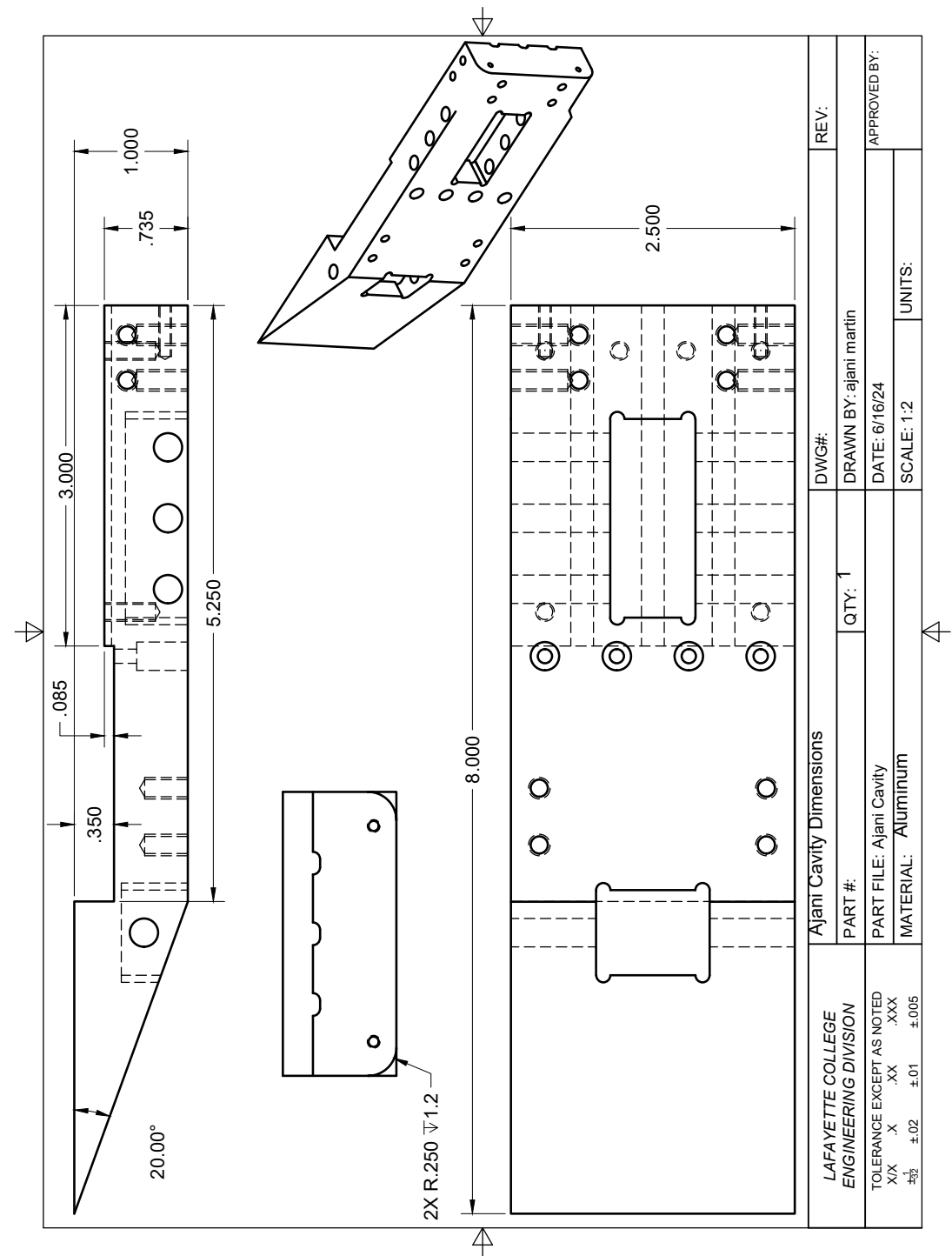
In conclusion, injecting fuel into the cavity flame holder from distinct jets located at the bottom produced consistent combustion and flame stabilization. This is due to the jets being able to penetrate the flow of air rushing into the cavity and inject directly into the recirculation zones of the cavity. Injecting fuel through this method also had the advantage of fueling the flame with a constant stream of fresh fuel and air, further maintaining the flame's stability.

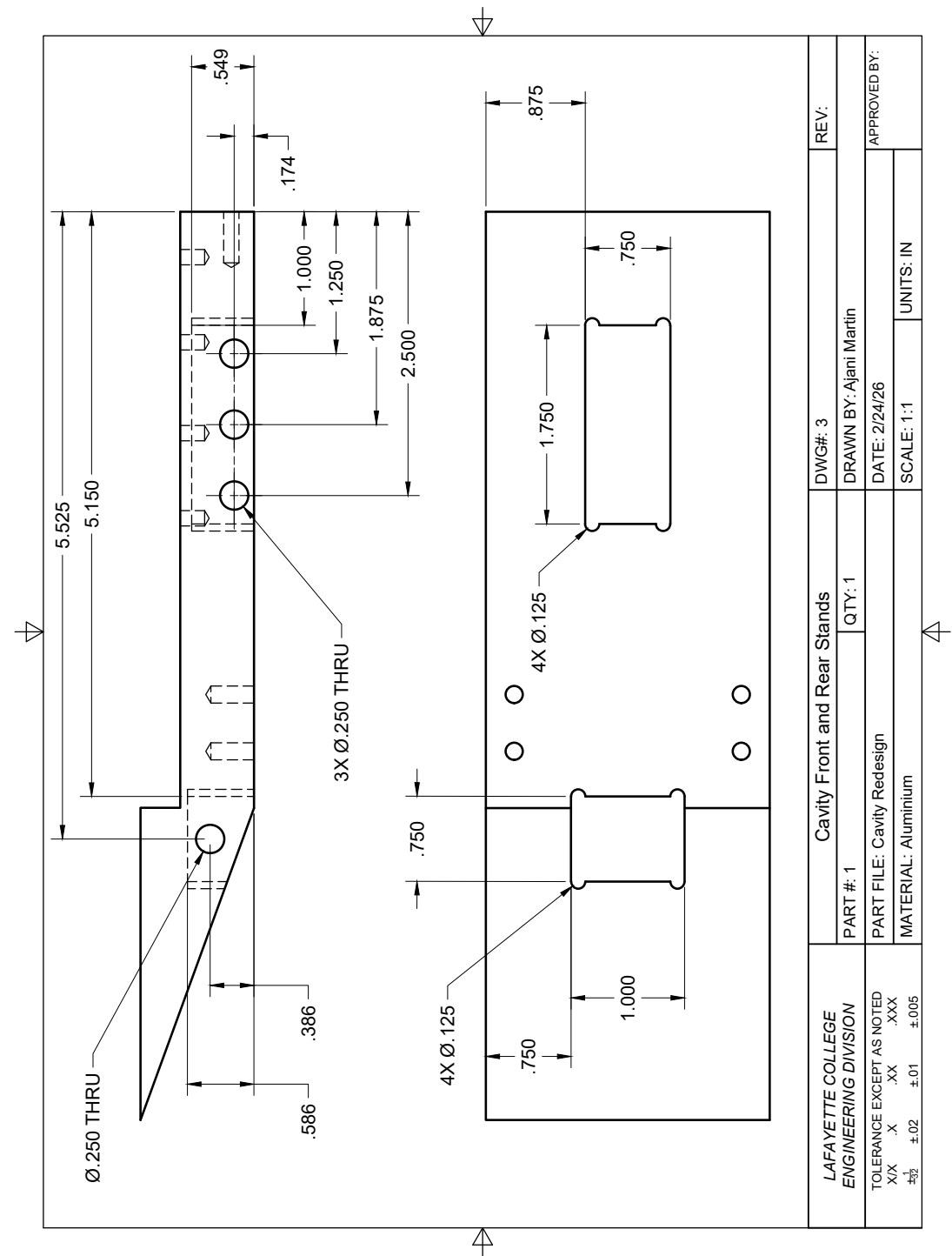
There are notably areas in which further testing in the Lafayette College Expansion Tube would significantly enhance and improve on the conclusions made on this thesis. First

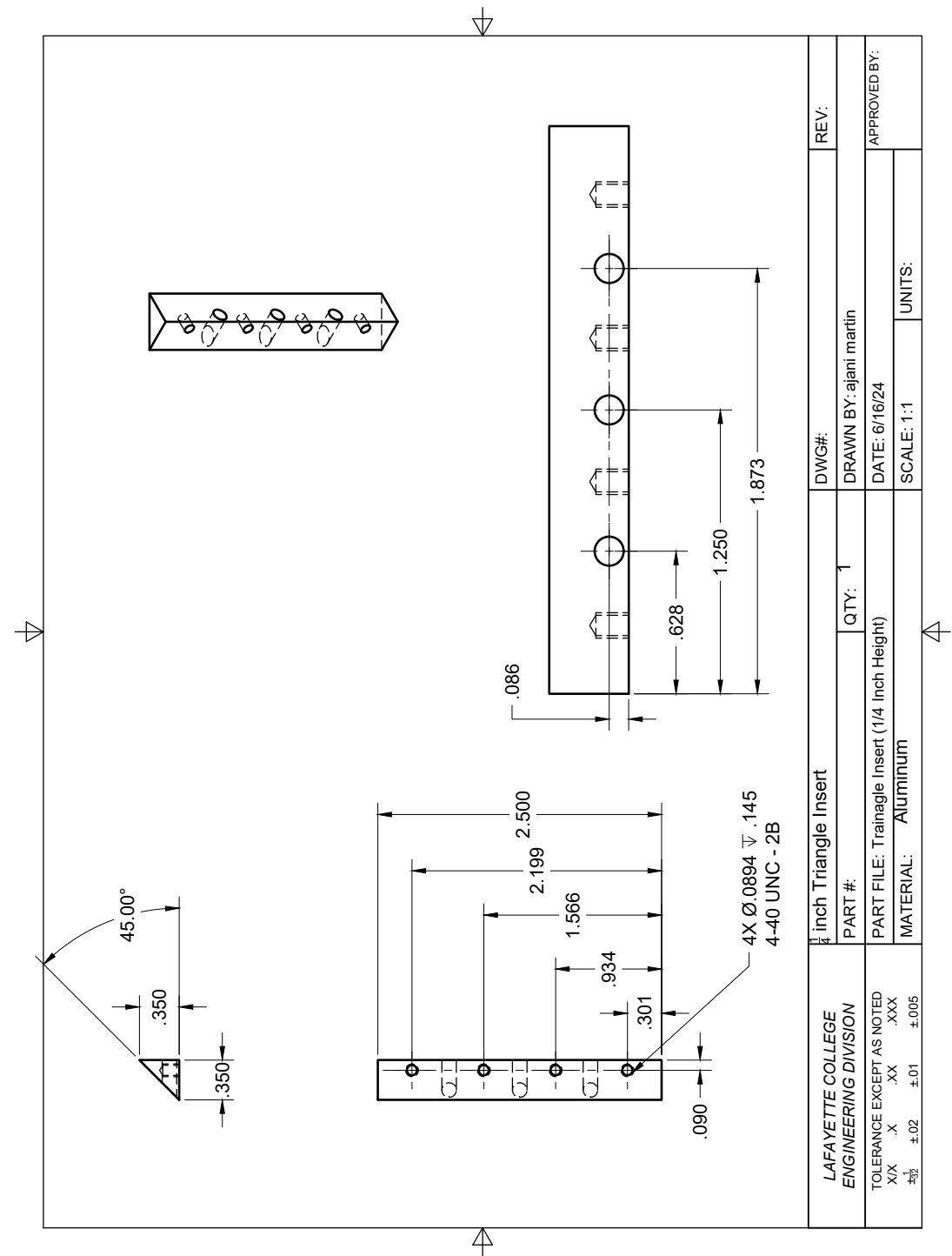
would be optimizing the injection pressure of the slit injection method. The cause of this method over-fuelling the cavity was due to the inherently high mass flow rate. The mass flow rate can be lowered by lowering the pressure flowing out of the ethylene canister, how the gauge regulating flow from the canister was incapable of going below 10psi. A low pressure gauge and a smaller cross sectional area for the slit injection method might produce more favorable results. Another future area that could be expanded on is combining the successful fueling locations as a means of optimizing the flame holding capability of the cavity. Previous students have found that fueling the cavity from small orifices from the floor of the cavity produced similar results to results gained from the three orifice fueling configuration discussed in this thesis. Combining these two methods might produce a cavity that both propagates combustion quicker and holds the flame longer. Finally, testing of both configurations could be altered by first finding the ideal mass flow for each configuration but altering the free stream parameters within the expansion tube. It is possible to find the free stream parameters that could limit combustion but not prevent it from happening entirely. This would allow for combustion to propagate but would make it less likely for over combustion to occur. Furthermore, holding free stream parameters as “constants” is not ideal since there may be large variations in the temperatures experienced by each experiment.

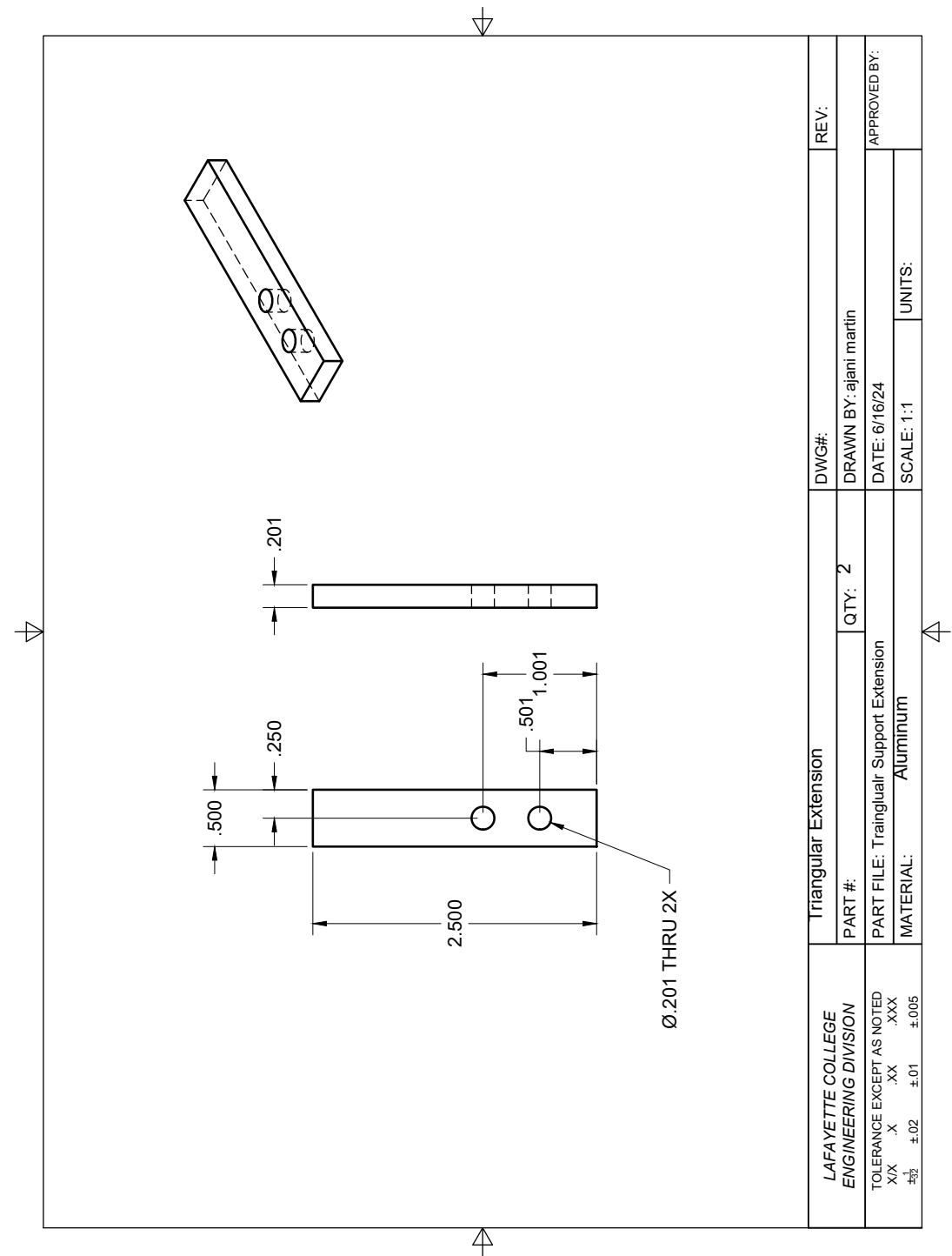
Appendix A

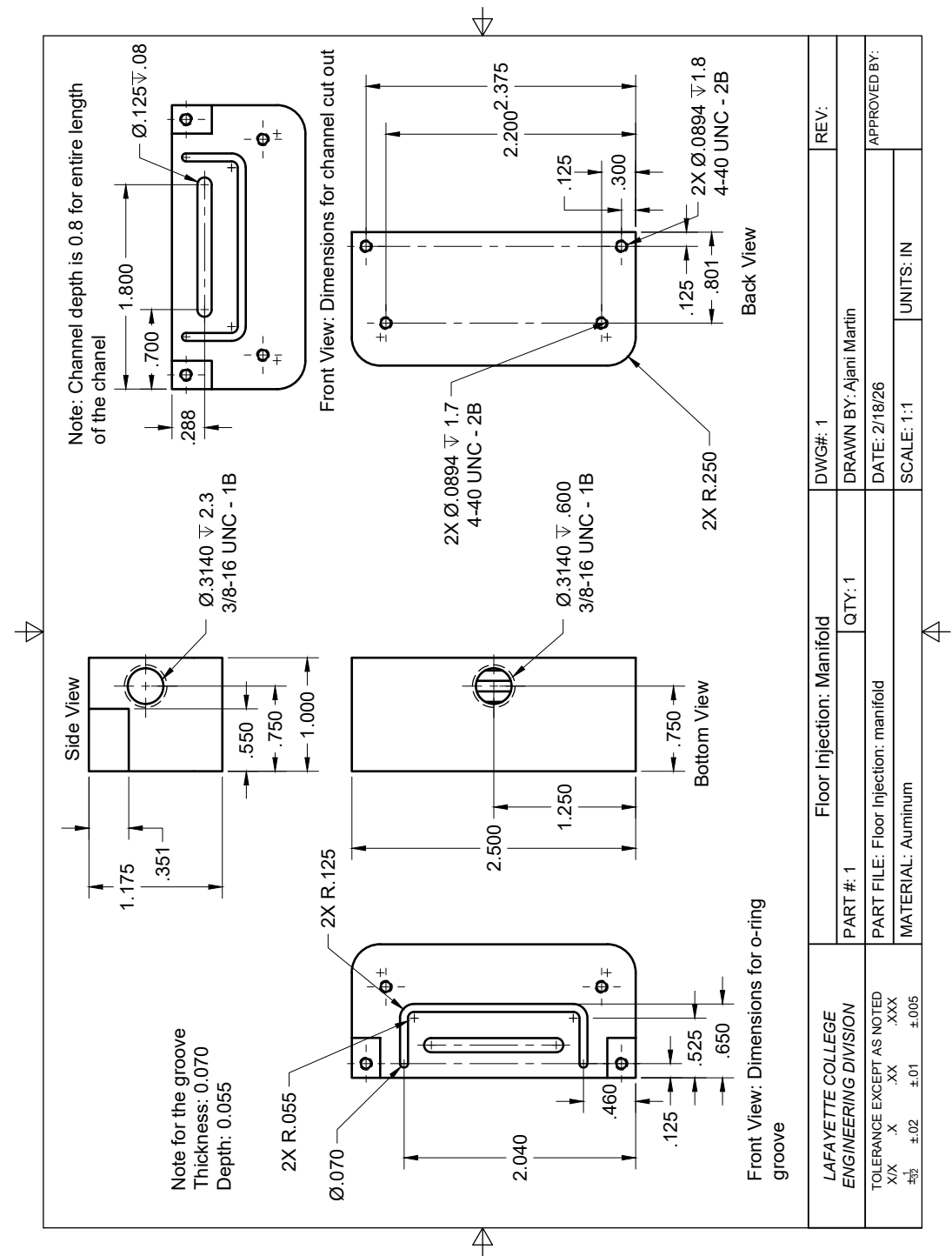
Part Drawings

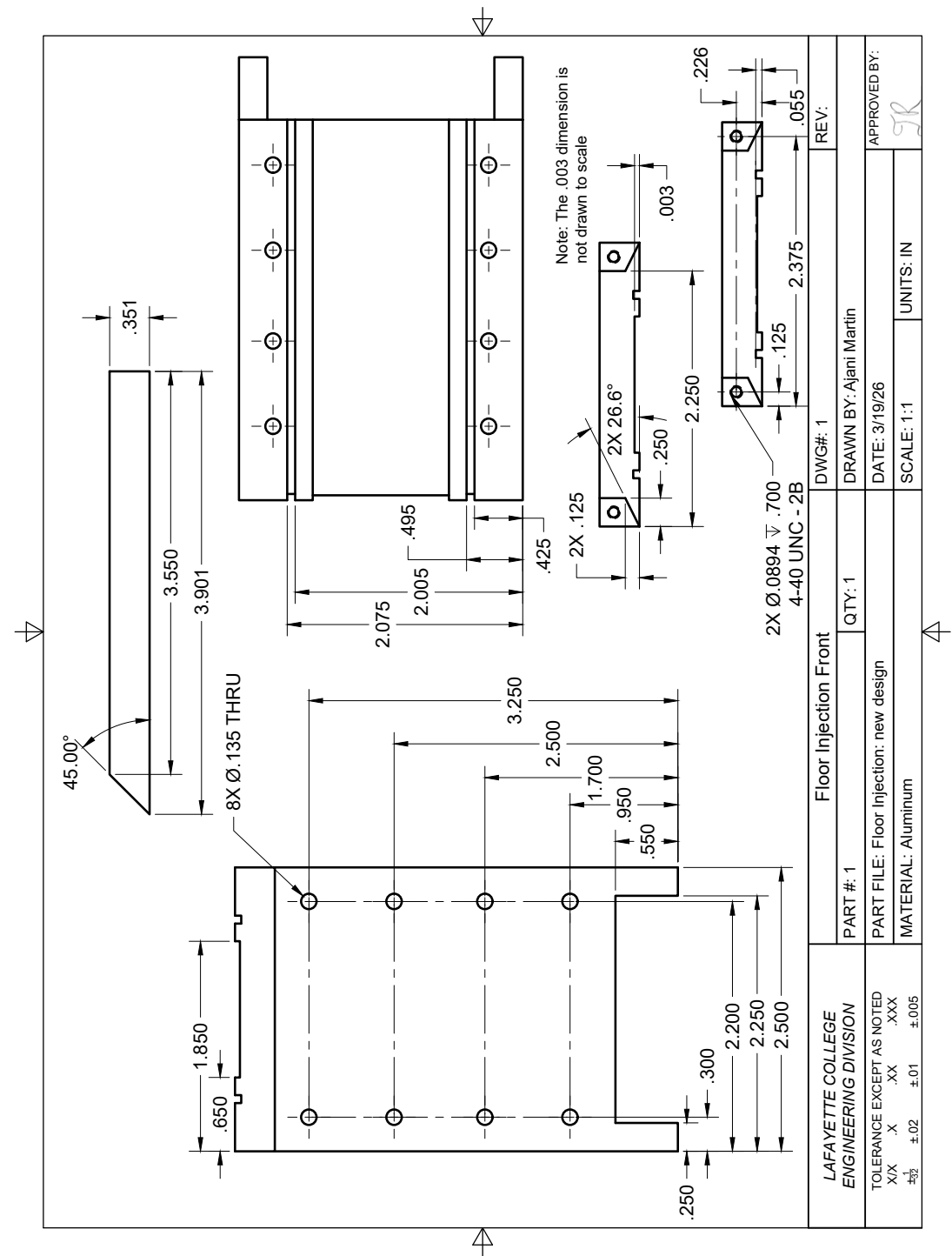












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